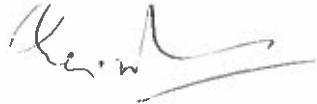


ANNUAL REPORT 1986



**NATIONAL ACADEMY OF
AGRICULTURAL RESEARCH MANAGEMENT**
Rajendranagar, Hyderabad-500 030 (A.P.)



ANNUAL REPORT 1986



**NATIONAL ACADEMY OF
AGRICULTURAL RESEARCH MANAGEMENT
Rajendranagr , Hyderabad-500 030 (A.P.)**

Compiled and edited by:

Dr.M.P.Singh, Professor
Sh.R.V.V.S.Prakasa Rao,
Asstt. Editor

Cover design and photography:

Sh. A. Murali Mohan Rao,
Artist
Sh. L. Venkateswarlu,
Photographer

Published by:

Dr. K.V. Raman, Director
NAARM, Rajendranagar,
Hyderabad.

Printed at:
Rajendranagar, Hyderabad.

NAARM, Offset Press

Cover page:

Administrative Block

P R E F A C E

The National Academy of Agricultural Research Management was set up in 1976 for providing training in management to various levels of scientists, administrators and transfer technology personnel employed in the Indian Council of Agricultural Research and the Agricultural Universities. The establishment of NAARM coincided with the development of progressive personnel policies of the Indian Council of Agricultural Research through the introduction of the Agricultural Research Service for the scientists cadre. The mandate of the Academy was later expanded to include training in Educational Technology and Management to assist development of a strong educational system in the country. The Academy has been planning, organizing and conducting training programmes, workshops and seminars in various aspects of Agricultural Research Management. So far over 3,000 scientists from ICAR Institutes and Agricultural Universities have undergone this training.

The Annual Report for the 1986, which is the first report that has been brought out by NAARM, gives an account of its programme and activities during the year. It also includes a brief description of the history and growth of NAARM as well as its structural reorganisation which was introduced in 1985. Recently an Advanced Centre on Agricultural Education and Research Management with FAO/UNDP/UNESCO support has also been established at the Academy.

The compilation and presentation of this report has been possible from the support received from all the faculty members. The preparation and editing of the report has been ably handled by Dr. Mahipal Singh. Suggestions for the improvement of presentation and contents of the Annual Report are welcome.


(K.V. Raman)
Director
NAARM, Hyderabad.

CONTENTS

Page No.

	Preface	
1.	INTRODUCTION	1
1.1.	Historical Background	1
1.2.	Objectives	2
2.	TRAINING PROGRAMMES	4
2.1.	Courses for Junior Level Agricultural Scientists and Development Workers	4
2.1.1.	Foundation Course on Agricultural Research Management (FCARM)	4
2.1.2.	Orientation Course on Agricultural Research Management (ARMOC)	10
2.2.	Courses for Middle Level and Senior Level Scientists	11
2.2.1.	Course on Agricultural Research Management (CARM)	11
2.2.2.	Course on Agricultural Research Project Management (CARPM)	11
2.2.3.	Course on Human Resources Management	12
2.2.4.	Course on Instructional Technology and Teaching Methods	12
2.3.	Summer Institutes	13
2.4.	Workshops and Seminars	13
2.5.	Consultancy Services	13

3.	STRUCTURAL ORGANISATION AND SALIENT ACTIVITIES	14
3.1.	Functional Units	14
3.1.1.	Agricultural Research Systems and Policies Unit	14
3.1.2.	Educational Systems and Technology Unit	22
3.1.3.	Transfer of Technology Systems and Policies Unit	30
3.1.4.	Project Management Unit	31
3.1.5.	Human Resources Development Unit	34
3.1.6.	Information and Documenta- tion Unit	37
3.1.7.	Centralised Services	41
	(i) Unit of Farm Operations and Management and the Agro- Technology Demonstration Blocks	41
	(ii) Production and Publications	45
	(iii) Administration and Accounts	46
3.2.	Staff Position	48
3.3.	Financial outlay	48
4.	UNDP Advanced Centre on Agricultural Education and Research Management	49
5.	Physical Facilities	51
5.1.	Closed Circuit Television System Lab	51
5.2.	Audio-visual laboratory including Photography Unit	51

5.3.	Mass Communication Laboratory	51
5.4.	Organizational Behaviour Laboratory	51
5.5.	Computer Facilities	51
5.6.	Bank Counter	52
6.	NAARM ALUMNI ASSOCIATION	53
7.	PUBLICATIONS	55
7.1.	Scientific papers published	55
7.2.	Books, Bulletins, Monographs, Technical Reports and Notes	55
7.3.	Papers presented at National and International Symposium/ Seminar/Conference/Workshops	55
7.4.	Special lectures delivered	56
8.	MISCELLANEOUS	60
8.1.	Faculty Development	60
8.2.	Foreign visits	60
8.3.	Distinguished visitors	61
8.4.	Honours and Awards	62
8.5.	Important Committees of the Academy	63
8.5.1.	Management Committee	63
8.5.2.	Steering Committee on Advanced Centre on Agricultural Education and Research Management	64
9.	PERSONNEL	66

1. INTRODUCTION

1.1. Historical background:

In the present era, training in Agricultural Research Management is increasingly becoming more important as science and technology is being harnessed for augmenting agricultural production in the country. Agriculture is gradually assuming the status of an organised industry. It has, thus, become important for all those concerned with agricultural research, education and development to be familiar with the principles of scientific management. The urgent need for an institutional set up to provide such training for workers engaged in agricultural research had long been felt and the Indian Council of Agricultural Research (ICAR) and the Government of India were seized of this matter in the early 1970s. In 1969-70, the Government of India was approached by the Food & Agricultural Organisation of the United Nations to host a training course in agricultural project analysis under the regional technical assistance project of the United Nations Development Programme (UNDP) and a Regional Training Course was organised at the Indian Agricultural Research Institute, New Delhi during November 2-28, 1970. This led to the setting up of National Training Centre for Agricultural Development and Project Planning at the IARI, New Delhi in April, 1971 as a part of the Fourth Five Year Plan activity. As the scheme was a self-liquidating one, at the end of the Fourth Five Year Plan, the activities of the National Training Centre were terminated on 28th February, 1974. However, a continuation of the National Training Centre work was found desirable by the authorities.

During the Fifth Five Year Plan, the ICAR took the initiative and appointed a high powered working group under the Chairmanship of Dr. George Jacob, Chairman of University Grants Commission, to give guidelines for formulating concrete proposals for organising a Central Staff College for Agriculture to ensure continuance of the activities of the National Training Centre. The working group had representatives not only from the concerned departments of the Government of India, but also other related educational agencies. The group identified the two main objectives of the proposed Central Staff College for Agriculture as (i) to function as a repository of ideas and information, both national and international, in the field of agricultural research and development and act as a clearing house for dissemination and utilisation of such information and know-how; and (ii) to undertake a programme of publishing high quality training material and allied publications as important and priority areas of action. The ICAR appointed a sub-committee of experts of the above

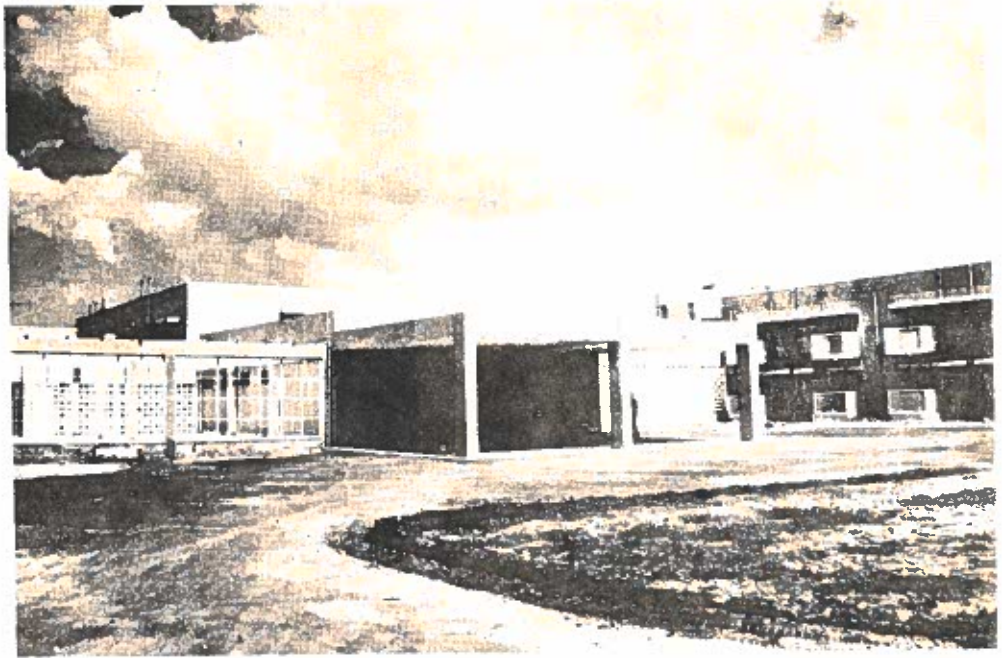
working group under the chairmanship of Prof. M.V.Mathur in 1975 to prepare details of the proposal. The sub-committee spelt out the need for the proposed Central Staff College for Agriculture with a fairly wide mandate and objectives.

With an outlay of Rs.50 lakhs for the Fifth Five Year Plan period, the Central Staff College for Agriculture came into existence on 1st September 1976. Later the Staff College was renamed as National Academy of Agricultural Research Management with effect from April 1, 1979.

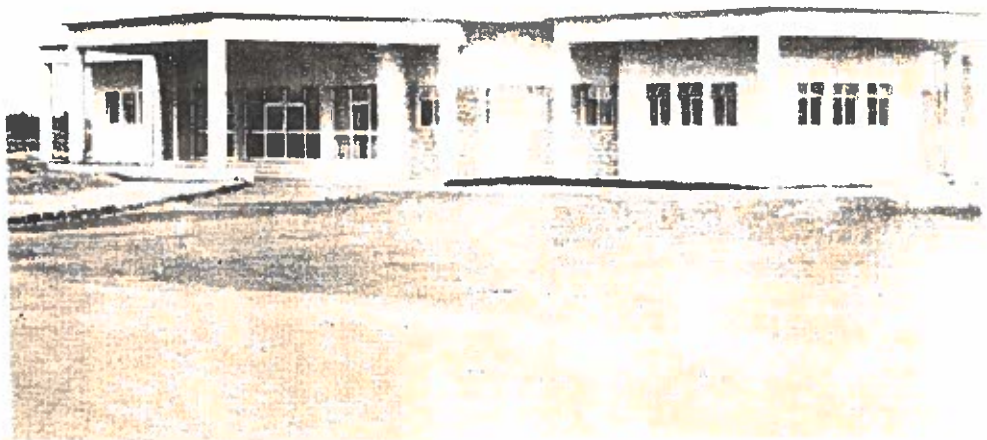
The Academy was started as a temporary expedient at Rajendranagar, Hyderabad where the Andhra Pradesh Agricultural University (APAU) placed at the disposal of the ICAR certain physical facilities such as hostel, class rooms and the administrative buildings. Subsequently, a broad based expert committee was appointed by the ICAR for recommending the permanent location of the Academy. The Committee visited several sites and locations and recommended Hyderabad as the permanent location since several infrastructure facilities and supporting institutions are available. For establishing the Academy, APAU released 50 hectares of land on lease for 99 years. On this land the Academy has constructed an Administrative Building, a Conference Hall, a Committee Room, an Academic Block, a 154 rooms Residence Hall, a Scientists Home and quarters for all categories of staff. The Academy is located at Rajendranagar and is 15 kms away from the centre of Hyderabad city. The campus is spread over 50 ha. of undulating picturesque landscape. The area is surrounded by APAU farms on Eastern, Northern and Southern sides and by National Institute of Rural Development on the Western side and is situated within Rajendranagar Municipal limits.

1.2. Objectives

The National Academy of Agricultural Research Management (NAARM) has been established with the main objectives of familiarising all those concerned with the Agricultural Research, Education and Development with the principles of scientific management in agricultural research and education. Since its inception it has taken up the responsibility to plan, organise and conduct training programmes, workshops and seminars in the various areas related to agricultural research management for the research scientists of the ICAR and Agricultural Universities. It has also been functioning as repository of ideas and information both at national and international level and to act as a clearing house for dissemination and utilisation of such information and know-how. The Academy has also recently taken up the training programmes in the area of educa-



Halls of Residence



Academic Block

tional technology, curriculum development and structured learning. It has also taken up research programmes and case studies in various fields of agricultural research management and educational technology.

The main objectives of NAARM are as follows:

1. To organise and conduct training programmes in Agricultural Research Management to Scientists at various levels.
2. To develop high quality resource material in the subject of Agricultural Research Management based on actual field experience.
3. To undertake systematic review and study of management problems of ICAR Research Institutes; Agricultural Universities, Agricultural Research Projects and Systems.
4. To organise, liaise and coordinate programmes of international cooperation in the field of Agricultural Research Management.
5. To plan, organise and conduct training programmes, workshops and seminars in educational technology, curriculum development and structured learning for the faculty of Agricultural Universities and ICAR Institutes.

The National Academy of Agricultural Research Management functions with a wide mandate. It is a novel, vibrant and emerging institution with a visionary outlook which has a dynamic approach and pragmatic policies with realistic expectations. It functions with a zest, zealously and commitment of purpose to fulfil the mandate of the Indian Council of Agricultural Research as the major national coordinating body for agricultural research and education, and thus fulfil the national aspiration for better management of science and agriculture.

2. TRAINING PROGRAMMES

The Academy has been planning, organising and conducting various types of training programmes to meet the need of the clientele groups working at different levels of management in the agricultural research and development organisations. Mainly three levels of clientele have been notified for training purposes, Junior, Middle and Senior.

2.1. Courses for Junior Level Agricultural Scientists and Development Workers

Two types of courses have been designed and are being offered. These two courses are:

2.1.1. Foundation Course on Agricultural Research Management (FCARM)

The Foundation Course for the scientists recruited to the Agricultural Research Service by the ICAR is one of the regular courses offered by the Academy. This Course aims at providing them exposure to concepts and principles of project management with special emphasis on project formulation and implementation as well as instruction in related areas such as group dynamics, interpersonal relations, interdisciplinary project formulation, motivation management and such other topics as would make them a more effective research workers. The Course is organised within the frame work of the following objectives.

- To acquaint the scientists with the general Agricultural Scenario in the country with reference to Research and Development.
- To present an insight to the rural society and structure.
- To equip them with the techniques in Research Project Management.
- To provide skills in organisational behaviour for high motivation, team work, conflict resolution etc.
- To provide training in agricultural communication and transfer of technology.
- To orient the scientists to the Indian Council of Agricultural Research and Agricultural Research Service.
- To foster a feeling of fraternity among the members of the ARS serving in different parts of the country.

The total duration of the course is five months. The trainees undergo six weeks training (First Phase) at the Academy oriented to the basics of Agricultural Research Management. After this training they spend about two months (Second Phase) at a Regional Research Station of National Agricultural Research Projects or in ICAR Institutes where they receive practical training in project identification and formulation and develop an appreciation for rural life and environment. They return to the Academy for a further programme of six weeks (Third Phase) where they get exposure to areas which will help them to become better research workers. During this training programme the trainees are exposed mainly to the subjects like Agricultural Research Management; Project Management, Organisational behaviour, Communication, Transfer of Technology, Administrative Rules and Procedures and Practical Classes on Audio-Visual Aids, Photography, Research Project Formulation, Indexing etc.

Agricultural Research Service (ARS) Scientists recruited through ASRB examination numbering 206 scientists (XXI to XXIV FCARM batch) were given intensive training in various aspects of Research Project Management viz. Research Programme Planning, Appraisal, Programme Evaluation and Review Technique, Critical Path Method, Conflict Management etc. Each participant was given training in the preparation and presentation of research projects on the basis of the information collected and training received during the Field Experience Training.

Course contents

The following Course contents were covered during the training.

I. General Agricultural Scenario in the country

- NAARM and its role
- Agro-ecological zones and cropping systems in the country
- Economics of Indian Agriculture
- Agricultural Prices
- Land Systems and measures of reform
- Agricultural Production - potential and constraints

II. Rural Society and Structure

- Rural development - philosophy and programme
- Characteristics of Indian rural life in comparison to urban life
- Rural leadership and rural development programmes
- Socio-economic surveys
- Tools of data collection for rural socio-economic surveys
- Preparation of viable research projects based on socio-economic surveys.

III. Research Project

- Agricultural Research Management Process
- Creativity and Problem solving
- Research Programme Planning
- The Research Project
- Project Appraisal
- Project Selection Exercise
- Project Management Techniques - CPM & PERT
- Resource Smoothing
- Critical Path Index
- Uncertainties in Research
- Management of Research Stations

IV. Development, Transfer and Adoption of Technology

- Technology development process - promoters and retarders
- Scientists and Transfer of Technology

- Institutional Structures for Transfer of Technology
- Treatment of Research Findings
- Semantics in Scientific Communication
- Communication through Audio-Visual Aids
- Agricultural communication through A.I.R.
- Agricultural communication through Doordarshan.

V. Histry, Structure and Organisation of Agricultural Research

- History and growth of ICAR
- Structure and organisation of ICAR
- Agricultural Research Service
- National Agricultural Research Project
- Linkages and coordination between ICAR and Agricultural Universities. Role of ICAR
- Linkages of ICAR with other National Research organisations
- Research Systems in Agricultural Universities
- Farming Systems Research
- All India Coordinated Research Projects of ICAR
- Constraint Analysis in Research and Adoption.

VI. Team Work and Interpersonal relationships

- Individual and the organisation
- Groups in Organisations
- Team Work
- Group Dynamics

- Motivation and Behaviour
- Motivation and Work
- Transactional Analysis
- Conflicts in Organisation
- Fatalism or positive thinking
- Case Analysis for Problem solving and decision making.

VII. Training in peripheral areas

- Communication skills
- Rapid reading and comprehension
- Writing of Scientific Paper
- Basics of computer programmes and use of computer in Agricultural Research
- Programming in BASIC and FORTRAN
- Energy Management in Agriculture
- Solar and Wind Energy
- Biomass and Energy
- Pollution
- Basic concepts of Linear Programming

VIII. Transfer of Technology

- Adoption of technology - stimulants and constraints for adoption
- Development communication in Agriculture
- Transfer of Technology Experience of Developed and Developing countries
- Rural Development Programmes in India.

IX. Organisation of Research

- Agricultural Research Systems in the World
- International Research Organisations
- Research Planning and coordination in ICAR
- Contribution of ICAR to Agricultural Education Systems in the country
- ICAR and International Collaboration in research.

X. Information and Library

- Source of Agricultural Information
- Citation Indexing
- Management of personal documents
- Book reviewing in the sciences
- Review literature
- Proof reading
- Poster sessions
- Technical Report Writing

XI. Trends in Research Work on some important crops/areas

- Crop Sciences
- Oilseeds Research
- Dryland Agriculture
- Pest Management
- Soil Management
- Fisheries Management
- Animal Management
- Horticultural Crops.

XII. Administration and Accounts

- General Service conditions - An Overview
- Stores - Management and Purchase Procedures
- Project budgeting and monitoring.

2.1.2. Orientation Course in Agricultural Research Management (ARMOC)

An Orientation Course in Agricultural Research Management (ARMOC) of one month duration is being conducted for the scientists working in various institutes of the ICAR. The objectives of the Course are as follows:

1. To discuss and acquaint with the issues relating to agricultural research management process.
2. To equip and acquaint with the techniques to do the work better as researcher.
3. To instil the philosophy of living, thinking and working together and foster a feeling of fraternity among the members of the ARS.

The topics in this Course broadly cover the Management Process, Research Project Management; Quantitative Techniques; Organisational behaviour, Management of Information System and Transfer of Technology.

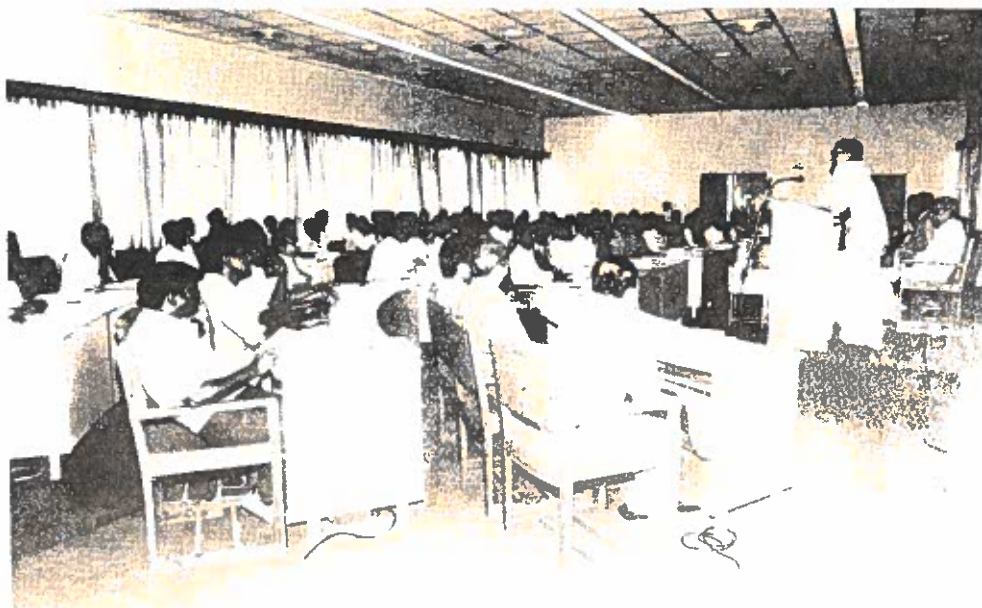
Two orientation courses on Agricultural Research Management for 94 S-1 inductees and promotees from S to S-1 were conducted during the year. The participants were exposed to various techniques of Agricultural Research Management.

Course Content

The following course contents were covered during the training:

- Agricultural Research Management
- Project Management
- Organisational Behaviour
- Communication
- Transfer of Technology

Inauguration of Training Programme



Inauguration of the
Foundation Week
Celebrations
(September 2-6, 1985)

2.2. Courses for Middle Level and Senior Level Scientists

2.2.1. Course on Agricultural Research Management (CARM)

It has been realised that the fulfilment of the national need for increased agricultural production requires that the Agricultural Scientists should organise and orient their efforts to rapid achievements of national goals, to utilise problem-centred rather than discipline-centred approaches to agricultural research and to take the initiative in arranging cooperation with all other agencies concerned with the agricultural development. Agricultural Research should be seen in developmental frame work including not only basic and applied investigations but the entire process of development of appropriate technologies and their wide spread dissemination.

The Academy is organising this Course on Agricultural Research Management (CARM) to increase effectiveness of each participant in his present position and prepare him for greater managerial responsibility by apprising him of the various aspects of research project management techniques and behavioural skills. The Course is based on the linkages between two groups of inter-related functions i.e. the functions of the agricultural research institutions and functions of a research manager.

2.2.2. Course on Agricultural Research Project Management (CARPM)

The importance of Agricultural Research in contributing to increased productivity is being fast recognised. Self-reliance and self-sufficiency are the main thrusts in agricultural production. Research and Development is to be geared up to provide technical service to the production units and help the farming community to produce more with economically feasible inputs. The research project managers have to formulate programmes, plan the projects, implement them, monitor and revise them systematically and scientifically. Appropriate linkages and communication channels with other units/divisions within and outside the research organisation to facilitate transfer of appropriate technology to the clients/farming community are necessary. An understanding of the above factors helps the research project leaders to develop a pragmatic approach to problems and make them more effective members of research team.

The Academy is organising this course on Agricultural Research Project Management (CARPM) to expose the research project managers in the ICAR Institutes to the concepts of research projects and application of the 'management approach' to the problems of

agricultural research units. The course is designed to serve the interests of senior research managers heading a division in the ICAR Institutes in the cadre of S3/S4 and the duration of the course is about two weeks.

2.2.3. Course on Human Resources Management

Agricultural Research has played a key role in increasing the food production threefold from land, water and other sources in India in the past two decades which resulted in Green, White and Blue revolutions. India is not only self-sufficient in foodgrains now but is in a position to export agricultural products to other countries. India has the largest manpower of Agricultural Scientists in the world, about 6480 in the ICAR system and 24,000 in the Agricultural Universities supported by an equally large technical staff. For optimum output, this group has to be properly motivated and their work is to be coordinated for achieving the objectives. This involves the management of human resources (scientists, technical and supporting staff) besides financial resources. Day by day Agricultural Research is becoming more specialised, complex and sophisticated. So planning and development of human resources for this purpose is very important.

The Academy is organising the Course on Human Resources Management to expose the Research Managers to the concept of human behaviour in organisations (Organisational Behaviour) and attempt to address the issues for human resources management in Agricultural Research. The course is designed to serve the interests of the Research Managers and Senior Scientists heading a division in the ICAR Institutes in the cadres of S-3/S-4 and the duration of the course is two weeks.

2.2.4. Course on Instructional Technology and Teaching Methods

Teachers today are facing the challenges on the modes of designing instructional systems by which conditions could be deliberately created to enable the learners achieve the desired educational goals in a specified manner. Teachers, as Educational leaders and Decision Makers, influence the students as well as subject matter. In addition to having the command on subject matter, teachers should also be familiar with the broad spectrum of instructional methodology, techniques and approaches that can be easily applied in the class room. The teacher should be able to analyse the educational training needs, keeping in view, the predetermined goals and working out instructional objectives, developing lesson plans, designing instructional material and teaching modes to be used and finalisation of suitable monitoring and evaluation techniques.

The Academy has recently started this Course with a view to expose the teachers to the need for appropriate planning of instructional strategies and optimum use of instructional resources keeping in view the complexities of teaching learning process coupled with the increase in number of students, the various educational streams from where they came as well as ever expanding horizons of modern knowledge.

This course on Instructional Technology and Teaching Methods is conducted for the senior faculty members of Agricultural Universities/ICAR Institutes who have interest in instructional technology and production of instructional aids. The duration of the course is two weeks.

2.3. Summer Institutes

The Academy organises Summer Institutes in the field of Agricultural Research Management, Educational and Instructional Technology for the benefit of the Faculty Members of the Agricultural Universities and ICAR Institutes. The Summer Institutes are usually of three weeks duration..

2.4. Workshops and Seminars

The different units of the Academy are planning, organising and conducting various types of workshops and seminars related to specialised topics in their own area of specialisation. The workshops and seminars are conducted on the basis of the research studies undertaken by them. The case materials and reading materials are also developed in advance for the benefit of the participants of workshops and seminars.

2.5. Consultancy Services

The Academy has been providing consultancy service in the area of Agricultural Research Management and Educational Technology. The Academy has been helping Agricultural Universities and ICAR Institutes from time to time in conducting training programmes in these areas. The Academy's Faculty are specialised in various sub-disciplines and deliver special lectures in the Institutes who arrange training programmes to their own staff. Institutes have found these lectures useful and periodically call on the services of the Faculty for this purpose.

3. STRUCTURAL ORGANISATION AND SALIENT ACTIVITIES

3.1. For the purpose of imparting effective training and undertake research studies in a meaningful way, the Academy is organised into six subject matter oriented functional units with a strong group of centralised services, as under:

3.1.1. Agricultural Research Systems & Policies

3.1.2. Educational Systems and Technology

3.1.3. Transfer of Technology Systems & Policies

3.1.4. Project Management

3.1.5. Human Resource Development

3.1.6. Information and Documentation

3.1.7. Centralised Services

(i) Farm Services and Agro-Technology Blocks

(ii) Production and Publications

(iii) Administration and Accounts.

All the above mentioned units have been conducting training programmes for Agricultural Service Personnel, Agricultural Scientists from Research Institutes and Agricultural Universities, Administrators and Technology Transfer personnel. The Organisational Chart of NAARM is given in Fig. 1.

3.1.1. Agricultural Research Systems & Policies Unit

Background/Introduction:

Research system is the backbone for agricultural development. It is the technology generated and information gathered and principles and procedures evolved by research system that are utilised by the other agricultural systems like education, training, extension, client and support. How the various research systems function and their effectiveness of functioning are of utmost importance. The Agricultural Research Systems & Policies Unit studies the various systems of agricultural research in order to strengthen the various research systems that are essential for agricultural development.

Objectives, Tasks and Functions:

1. To conduct systematic research in the areas of Agricultural Research Systems and Policies.
2. To organise specialised training programme in systems approach with regard to specific problems of Agricultural Research Management. A/
3. To evolve a package of empirically proven, reliable and valid research policies to be recommended for implementation by the concerned authorities.
4. To offer consultancy advisory services to the various research agencies for effective functioning of various research systems in areas like manpower planning and utilisation etc.

Activities during the year 1986

1. Review of Indo-US Agricultural Research Project

The study was undertaken to review the progress made under Indo-US agricultural research project and to identify the constraints in the implementation and to suggest solutions for the Indo-US agricultural research project implementation.

In the review, the efforts were focussed on the following three sub-projects which are now being implemented as per the Project Grant Agreement.

- a. Soybean Processing and Utilisation (SPU), operating at CIAE, Bhopal and G.B.Pant University of Agriculture and Technology, Pantnagar.
- b. Post-harvest Technology of Fruits and Vegetables (PHT-FV) operating at IARI-New Delhi, IIHR, Bangalore, and CIHNP, Lucknow.
- c. Project Implementation Unit (PIU), operating at the ICAR Headquarters.

The study has revealed many constraints which the Indo-US Agricultural Research Project has faced during the first two years of its implementation. At the sub-projects development stage, some of the problems the project has faced are:

- i) considerable delays in the constitution of the design teams and sub-project design;

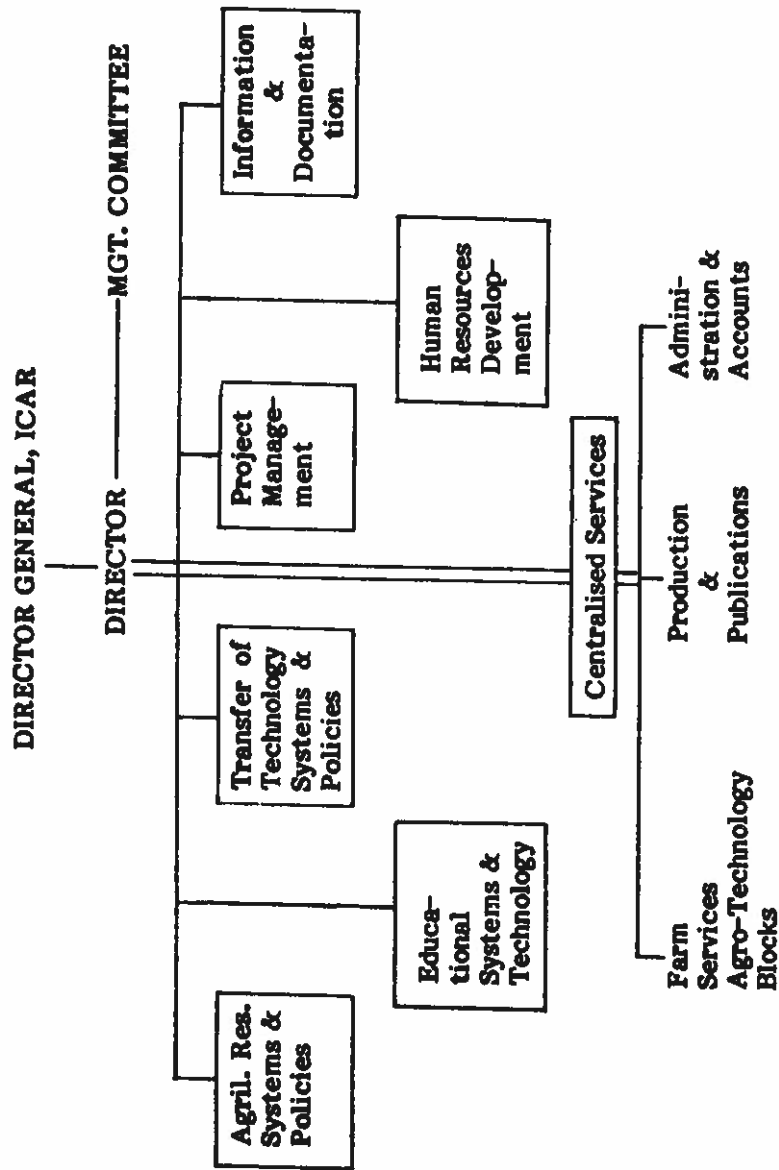
- ii) limited representation of the concerned scientists leading to less commitment; and
- iii) delay in the sanctioning of the sub-projects owing to the existing lengthy administrative procedures of scrutiny by different agencies such as ICAR, DARE and DEA. Some of the problems associated with technical formulation of the sub-projects are:
 - a. delays in the constitution of Management Teams and in arranging their visits to US for identifying centres for training of Indian scientists, inviting consultants and selecting equipments for import;
 - b. Inevitable delays in grounding the Project by way of recruiting staff, creating infrastructural facilities immediately after the formal approval and sanctioning of the sub-projects;
 - c. insufficient understanding and appreciation of the objective of the Project at various levels;
 - d. difficulties faced in obtaining the proper placement of training due to incomplete and inadequate information provided; and
 - e. delays in getting clearance of proposals for training scientists abroad due to lengthy administrative procedures.

The major problem associated with the role clarity and line control identified is: the roles and functions of PIU, Subject Matter Division and DARE (IC Division) at the ICAR Headquarters are not clearly understood and appreciated by the persons concerned. Some of the problems at the implementing centres level are (i) delay in training the Indian scientists abroad; (ii) delay in the arrival of consultants; (iii) delay in the import of equipments for want of timely clearance of NMIC; (iv) difficulties in reflecting the cost of imported equipments in planned budget; (v) non-availability of the full complement of the staff at different centres; (vi) confusion among the deployed scientists about their role and privileges in the sub-projects; (vii) timely non-release of funds; (viii) lack of proper monitoring mechanism at various levels; and (ix) the need for ensuring proper organisational climate at the centres level for effective implementation.

(Dr.T.Balaguru)

Fig.1:

ORGANISATIONAL STRUCTURE



2. Study on Scientific Manpower Planning

The Unit has taken up a Project on Scientific Manpower Planning in ICAR Research Institutes. The overall objective of the project is to study the scientific manpower planning in ICAR Institutes with special reference to utilisation of the present scientific manpower in achieving the set objectives of the Institute and to predict the future manpower requirement for selected institutes. Ten ICAR Institutes have been selected for the purpose of the study and questionnaire has been prepared to get information with regard to manpower in different divisions of the selected ten ICAR Institutes. Processing of data received is in progress.

(Mr.V.E.Sabarathnam and Dr.M.Rajagopalan)

3. Study on the trends in agricultural research and policies in India

This project is to study the trends in the research and development policies in India and how the development needs are translated into research policies. The study will be undertaken from secondary source data like Plan documents and interviews with the persons concerned. Relevant information has been collected and is being analysed.

(Dr.T.Balaguru & Dr.M.Rajagopalan)

4. A comparative study on the Agricultural Research Systems of the World

The study is to examine the various agricultural research systems that are followed in different countries including India and to suggest improvements. Data have been collected and are being analysed.

(Dr.T.Balaguru & Dr.M. Rajagopalan)

5. Exposure of Scientists on the use of computers

The S-1 Scientist trainees were given the exposure on the use of computers with the help of micro computer UPTRON S-850 installed at the Academy. The Scientists were first given exposure from the users point of view on the basic aspects of computer use and the use of programmes in the two computer languages, BASIC & FORTRAN. Then they were given an opportunity of operating the computer individually to solve some of the research and management problems using the programmes already developed by the Unit.

Course material developed

The following course materials were developed and delivered by the Unit to various courses, Summer Institutes and Workshops conducted during the year 1986 at NAARM.

1. ICAR and International Collaboration
2. Research System in Agricultural Universities
3. Linkages and Coordination with ICAR and Agricultural Universities
4. Linkages of ICAR with other National Research Organisations.
5. Research Management Process
6. Linear Programming
7. Sample Survey Methods
8. Sources of bias in Data Collection
9. Resource Smoothing
10. Decision Networks
11. Uncertainties in Research
12. Technology Development Process in Agriculture
13. Treatment of Research findings in Research Systems
14. DOT (Development of Technology) Oriented Research Efficiency
15. Socio-economic survey for formulation of Agricultural Research Projects
16. Methods of Data Collection and Analysis
17. Preparation of viable Research Projects based on Socio-economic survey
18. Consequence of Agricultural Technology
19. Coding and Editing of Data.

20. Role of Simulation games in Education and Training
21. Agro-ecological zones and trends in Agricultural Production
22. Basics of Computers
23. Programming in BASIC
24. Green Revolution Game.

Workshops

Workshop on Management of Indo-US Agricultural Research Project

A two-day workshop on Management of Indo-US Agricultural Research Project was organised from November 11-12, 1986 to review and to evaluate the progress made so far in the implementation of the Indo-US Agricultural Research Project and to suggest course of action to overcome constraints in the implementation. The Workshop was held under the chairmanship of Dr.N.S.Randhawa, Director-General, ICAR. Twenty four participants from various organisations and agencies like ICAR Headquarters, USAID, WINROCK International, Agricultural Universities and ICAR Institutes participated in the Workshop. Some of the important recommendations made for effective implementation of the project are as follows:

1. For speedier and effective development of the sub-projects, efforts should be made to constitute the Design Team as quickly as possible by concentrating on the priority areas already identified, and by associating the concerned working scientists for the development stage and by involving experts from ICAR Institutes or Agricultural Universities, wherever necessary.
2. For speeding up the sanctioning process, the possibility of authorising ICAR/DARE to directly contact the USAID for the sanctioning of individual sub-projects based upon the certificate of Guarantee on security and sensitivity issues by the Director-General (ICAR) as in the case of PL 480 schemes, once the Umbrella Project has been approved by the DEA may be explored. Similarly, only the Umbrella Project needs to be approved by the PIC and subsequently the PAMC may be authorised to approve the individual sub-projects.
3. For effective implementation, the Management Team consisting of the consultant who was in the Design Team apart from the scientists responsible for implementation should be constituted

within four weeks of the availability of the Design Team's Report and should visit US for identifying the institutions for training, consultants and equipments for import.

4. An initial one year gestation period subsequent for creating facilities and subsequent five years working period should be sanctioned for more meaningful implementation. At the same time, all formalities for the future sub-projects should be completed before June, 1987.
5. In order to orient the scientists on the special requirements and operational aspects of the Project, an Operational Manual on the lines of NARP should be prepared.
6. As far as the training of scientists is concerned, the Coordination Committee should finalise the names quickly soon after the Management Team's visit suggesting alternative institutions for facilitating the Winrock International. For speeding up the process of clearance, an annual training programme schedule may be prepared for getting clearance by the DEA at a time. The possibility of delegating the scrutiny work for individual sub-projects to the DARE by the DEA once the Umbrella Project has been approved by the latter. The training should be completed in a phased manner within the first 3 years and the deployed scientists should also be given this opportunity.
7. The consultants visiting the sub-projects should be as far as possible those involved in the Design Teams and their visits should be arranged in a phased manner throughout the life of the sub-projects and at a time when the trained scientists are in position.
8. As far as the import of equipments is concerned, the latest and complete specifications with alternatives should be provided for speeding the process and efforts should be made to get the NMIC clearance as quickly as possible. The foreign equipment imported could form an additionality to the planned budget or the cost can be met by the internal sources of ICAR.
9. To delineate the functions of PIU, SMD and DARE (IC Division), the committee constituted by the workshop came up with concrete recommendations for clarifying the roles of the concerned agencies.
10. Since the sub-projects are time-bound, full complement of the staff should be provided immediately through recruitment or deployment.

11. Timely release of funds should be accorded on an annual basis as it is the case with other schemes of the Institutes, and the PIU should be made responsible for regular monitoring of the financial transactions through computerised system.
12. For proper monitoring, monthly by the Principal Investigators at the implementing centre level, quarterly by the Institute Directors at the Institute level, half-yearly by the Coordination Committee, annually by SMD (Technical progress) and PIU (other aspects) at the ICAR Headquarters and regular annual review by ICAR (PAMC) and USAID must be ensured. Different monitoring proforma for this purpose have been evolved.
13. Proper organisational climate must be ensured by both the Institute Directors and the Principal Investigators for more meaningful implementation of the Project.

Consultancy and Advisory Service

A request was made to NAARM by Director-General, ICAR to undertake a study on the implementation and progress of the Indo-US Agricultural Research Project with the following terms of reference:

- (i) to review and to evaluate the progress made so far in the implementation of the Indo-US Agricultural Research Project.
- (ii) Identify constraints in the implementation; and
- (iii) Suggest course of action to overcome constraints.

The review study was undertaken by Dr.T.Balaguru under the overall guidance and supervision of Dr.K.V.Raman, Director and Dr.M.Rajagopalan, Professor, NAARM. The study was conducted in three phases - the first phase involved getting information from the sub-project implementation centres on a structured proforma, the second phase involved the visit of the NAARM faculty to these implementation centres, viz. ICAR Headquarters, USAID and WINROCK International Officials, Department of Economic Affairs, Department of Agricultural Research and Education and detailed discussions with the concerned officials, and the third phase involved the organisation of the workshop based on the review report as a result of the first two phases in order to arrive at specific recommendations.

Some of the suggestions made after critically analysing the constraints the Project has so far faced during implementation are: there is need to orient the scientists at various levels on the special

requirements and operational aspects of the Project by preparing an Operational Manual; (ii) it is necessary to provide one year gestation period for creating facilities before the sub-projects take off; (iii) the entire process of sub-projects design and clearance must be speeded up through the concerted efforts of various agencies involved; (iv) working scientists should be associated with the Design and Management Teams for ensuring their commitment to the Project; (v) the entire process of training Indian scientists abroad, import of equipment and inviting consultants must be speeded up by proper advance planning; (vi) the length of administrative procedure as far as the clearance of sub-projects, approval of scientists for training clearance of consultants' visits, import of equipment, need to be shortened by delegating authorities; (vii) the role of PIU, SMD and DARE (IC Division) need to be streamlined for effective implementation; (viii) timely release of funds should be ensured; (ix) proper monitoring mechanism should be evolved for various levels; (x) the full complement of staff should be provided; (xi) the apprehension on the part of the deployed scientists should be removed and they should be made part of the project and considered for all privileges due to them; (xii) proper organisational climate should be ensured so that the working scientists could involve in the project with full commitment and dedication.

3.1.2. Educational Systems and Technology Unit

Background/Introduction

In recent years India has made rapid strides in imparting higher agricultural education through a network of 25 Agricultural Universities, about 100 Agricultural Colleges and ICAR Institutes. Every year a large number of agricultural graduates and postgraduates are coming out of these Institutions. Though the Agricultural Universities have inherited and adopted the educational pattern of Land Grant University of U.S.A. i.e. the integration of teaching, research and extension education functions, it is felt that the instructional methods adopted in various Universities differ in their content and methods of imparting education and training. There is an urgent need to make higher agricultural education more meaningful and purposeful. It is realised that though the teachers are academically qualified, many of them follow the traditional methods of teaching which do not meet the demands of practical utilisation of knowledge gathered.

Currently the country is facing new challenges in the agricultural front owing to the sudden explosion in scientific knowledge and population and, in turn, to tune up the curriculum of agricultural education suitably. Hence, there is a need to study the issues relating

to the content, the structure and the function of the Agricultural Curriculum at the undergraduate and postgraduate levels of education offered in our Agricultural Universities and to explore the ways of making it more purposeful and responsive to the needs of changing time.

The primary function of this unit is to render assistance to Agricultural Universities and ICAR Institutes dealing with education, support for development of better programmes and better teaching techniques. Modern education technology interacts closely with subject matter specialists to develop viable and useful curricula courses and teaching techniques.

Objectives, Tasks and Functions

1. To conduct training in Educational Technology as applicable to curriculum.
2. Educational planning, administration of management.
3. Instruction techniques and material production.
4. Studies on teaching-learning process including evaluation.
5. Consultancy and Advisory services.

Activities during the year 1986

The following three courses were planned, organised and coordinated in the area of Research Management in Home Science and Communication.

1. Course on Effective Communication

A six day course on effective communication was conducted from March 17-22, 1986 for the benefit of senior scientists of ICAR. The Course was coordinated by Dr. Devesh Kishore, Professor and Unit-in-Charge. The topics covered in the course were: Communication Process - meaning, concept and communication models; Communication through Audio-Visual aids; Office Communication; Semantics in Scientific Communication; Communication in Organisation; Effective use of Audio-Visuals; Writing for Radio; Scientific Paper Writing; Rapid Reading and Comprehension; Television in Rural Communication; Reporting for Agriculture and Effective contribution of Review Papers.

2. Course on Research Management in Home Science

A Course on Research Management in Home Science was conducted from August 19-30, 1986 to expose the Professors and

Associate Professors in Home Science Colleges of Agricultural Universities and ICAR Institutes to the basic concepts of developing research projects, application of pragmatic and interdisciplinary management approach to the research problem so as to prepare them to be more effective members of research teams. The Course was coordinated by Dr. Devesh Kishore, Professor and Unit-in-Charge and the following topics were covered: Research Project Management; Research Methodology; Research Organisational Behaviour; Problem solving and Decision Making; Organisational Communication and Mass Communication.

Twenty five Faculty Members from 16 Institutions who are involved in development and management of research projects from all over the country participated in the course.

3. Course on Mass Communication

The Course on Mass Communication was conducted from 29th September to 1st October, 1986 for the benefit of the Senior Scientists and students of Postgraduate Diploma Course in Dairy Business Management of National Dairy Research Institute, Southern Regional Centre, Bangalore. The programme was coordinated by Dr. Devesh Kishore and the following topics were broadly covered: Organisational Communication; Communication; Mass Media and its role in dissemination of information; Decision Making Process; Report Writing and Techniques of Reviewing Books on Communication; Role of Television in Dissemination of Information and Mode of Writing Scripts.

Thirteen Senior Scientists and Scholars attended the Course.

Course material developed

1. Conceptual Framework and Guidelines for Curriculum Development in Agricultural Universities.
2. Agricultural Research Management Process
3. Television in Rural Communication
4. Mass Communication with special reference to rural broadcasting
5. Rural Broadcasting
6. Problem Solving and Decision Making
7. Organisational Communication

8. Ways and means of improving communication skills

9. Integrated Rural Development

10. Curriculum Design and Development.

Workshops

The following four Workshops in the area of Educational Systems and Instructional Technology were planned, organised and conducted for the benefit of Deans/Principals/Directors/Faculty Members of Agricultural Colleges, ICAR Institutes and Directors of Summer Institutes in the year 1986.

1. Workshop on Modification of Undergraduate Courses

A three-day Workshop on Modification of Undergraduate Courses was organised from January 8-10, 1986, on the special request of the Vice-Chancellor of Andhra Pradesh Agricultural University for the Faculty Members who are presently teaching undergraduate courses in the areas of Agronomy, Entomology, Plant Breeding and Genetics. The main objectives of the Workshop were: (i) to understand the basic process of course modification keeping in view the principles of educational technology; (ii) to modify course(s) presently taught at the University in relation to curriculum objectives and changing needs; and (iii) to identify suitable teaching strategies and instructional tools for each topic of the course.

During the Workshop sessions the different topics covered were: Formulation and Modification of Course(s); Presentation of Case studies on Course Modification; Modification of present course(s) taught by the participants and Identification of suitable teaching strategies and instructional tools.

The Workshop recommended that all the courses of the Department should be modified as per the procedure discussed during the Workshop and as per the exercise done by the participants for one course presently taught by them. The lesson plans for each topic of the course should also be worked out. Lesson Plans should be prepared for each topic of the course as per guidelines given during the Workshop and discussion held during the Summing up Session of lesson plans. (A Workshop should be organised in the area of Instructional Technology.)

Twelve Faculty Members from Andhra Pradesh Agricultural University participated in the Workshop.

2. Workshop on Curriculum Design and Development

A two day Workshop on Curriculum Design and Development was organised from April 9-10, 1986 to expose the Directors of the Summer Institutes to the basic principles of Educational Technology on Development of the Course and Curriculum. The objective of the Workshop was to make them aware of various latest techniques of instructions by which teaching-learning process can be made more effective. (Participants were explained the expectations regarding the developing of the course from the Summer Institutes.) During the Workshop sessions following topics were covered: Concept of Course Development; Framework and Guidelines for Curriculum Development; Case Study on Development of a Course and Translating Syllabus into teaching programmes.

The participants were told the basic concepts and philosophy of including the objectives of the Summer Institutes. The framework and guidelines for curriculum development were discussed threadbare. The case studies which were developed by the Academy on development of course were presented. This helped them in understanding the process of modification/developing a course. Scientific Information Management Aids in curriculum design and development were also explained. The participants felt that this Workshop has immensely helped them in understanding the basic concepts, principles and processes of Educational Technology and Instructional Techniques in development of the course.

Thirteen Directors of different Summer Institutes participated in the Workshop.

3. Workshop in Instructional Technology in Home Science

One day Workshop on Instructional Technology in Home Science was organised on September 20, 1986 for the benefit of all the Faculty Members of College of Home Science. The main objective of the Workshop was to expose the Faculty Members to the modern instructional systems and techniques. During the Workshop following topics were covered: Conceptual framework and guidelines for curriculum development; Formulation and modification of courses; Instructional design; Instructional methods and Role of Audio-visual aids in Teaching-learning process.

The Workshop recommended that a two weeks training programme should be organised on Educational Technology in Home Science. The participants should be given assignments in developing and modifying the curriculum at micro and macro levels. They should

be able to evolve suitable instructional methods suiting to their present teaching programmes at the University while attending the programme. The Faculty Members should be given opportunities to prepare the instructional material and audio-visual aids for making teaching-learning process effective.

Forty two participants from Home Science College, A.P. Agricultural University participated in the Workshop.

4. Workshop on Educational Technology in Home Science

A four day Workshop on Educational Technology in Home Science was organised from December 9-12, 1986 to create awareness and consciousness of the need for the scientific development of Home Science Curriculum in perspective of the total Higher Education System. During the Workshop the participants discussed the issues relating to improve the undergraduate and postgraduate curricula in relation to the developmental needs and plans of the country and the problems and constraints in relation to the implementation of the curricular changes and to work out effective and practical solutions.

During the Workshop sessions following topics were covered: Curriculum Development and Design; Formulation and Modification of courses; Approach to syllabus specification; Instructional Systems Design and Methods and Issues relating to Curriculum implementation.

The issues related to the curriculum implementation were discussed threadbare and following recommendations were made by the Workshop:

- i) The various Departments of Home Science Colleges should interact and integrate the undergraduate programme in such a way that proper implementation of the curriculum is made possible.
- ii) Orientation training programme on Curriculum Planning should be organised from time to time by the Home Science College of Agricultural University.
- iii) The participants felt that semester system should be preferred over other systems of curriculum implementation prevailing in different Universities such as Trimester or Annual or Traditional Systems.
- iv) The Departments offering various courses should be strengthened in terms of infrastructure facilities, such as audio-visual aids, well equiped laboratories and reprographic services.

The participants developed following curriculum keeping in view the principles of Educational Technology and Curriculum Development:

- i) Diploma curriculum in Home Science
- ii) Undergraduate curriculum in Home Science
- iii) Postgraduate curriculum in Home Science.

Fifteen participants from different Home Science Colleges of Agricultural Universities and General Universities participated in the Workshop.

Summer Institutes

Summer Institute on Educational Technology with special emphasis on Instructional Techniques

A Summer Institute on Educational Technology with special emphasis on Instructional Techniques was planned, organised and conducted from May 26 - June 14, 1986. The main objective of the Summer Institute was to expose the need for appropriate planning of instructional strategies and optimum use of instructional resources keeping in view the complexities of teaching-learning process coupled with the increase in number of students, the various educational streams from where they come as well as several expanding horizons of modern knowledge.

The following topics were covered in the Summer Institute: Educational Teaching; Role of Instructional Technology in Effective Teaching-Learning Process; Planning Learning Experiences; Trends approaches and Techniques of Instructional Technology; Techniques of Instructional Technology; Techniques of conducting a class effectively; Role of instructional aids in teaching, types of aids; Planning and selecting audio-visual aids, guidance in selecting and making instructional aids; Making practicals learning oriented and Programmed instruction and structured learning.

The participants of the Summer Institute strongly recommended that an Educational Technology Cell should be established at each Agricultural University and National Institutes of ICAR. Following should be the objectives of the Educational Technology Cell:

- To restructure and modify the existing curriculum in Agricultural Universities and ICAR Institutes to meet the general and specific needs of the learners.

- To bring uniformity in the course curriculum throughout the Agricultural Universities/ICAR Institutes.
- To systematize the teaching-learning process to the expected terminal behaviour.
- To update and improve the teaching standards.
- To delineate the constraints in the implementation of course curriculum and to develop suitable strategies to overcome these constraints.
- To impart short term training programmes to all University teachers/research workers/extension workers in educational technology.
- To develop resource material in the form of audio-visual aids which could be used by the teachers.
- To conduct research in agricultural educational technology.
- To acquaint the teaching community regarding latest innovations in educational technology such as simulation, role playing, structured learning, problem solving, etc. etc.

The Educational Technology Cell should organise short-term training programmes for the benefit of the Faculty in the following areas:

Development and modification of curriculum at macro and micro levels; Updating teaching and training methodologies and instructional techniques; Production of resource materials and instructional aids; and To conduct studies on more effective teaching methodologies and strategies, and developing low cost teaching aids.

The participants of Summer Institute on Educational Technology may serve as resource persons in establishing the cell at their respective Universities/Institutions. However, the Institutions like National Academy of Agricultural Research Management (NAARM), Hyderabad may provide a periodic updating of their skills in appropriate training. The cell should have facilities for production of audio-visual aids including production of transparencies, photographic slides etc. It should also have a CCTV Unit so that video cassettes and learning packages can be developed.

Twenty three Faculty Members from Agricultural Universities and ICAR Institutes participated in the Summer Institute.

Dr. Devesh Kishore was the Director of the Summer Institute.

Consultancy and Advisory Service

The consultancy and advisory services were rendered to the following institutions in the field of Educational Technology/Communication.

1. School of Communication Studies, University of Poona in developing Curriculum for the Advanced Courses in Communication.
2. National Dairy Research Institute, Karnal in effective implementation of the curriculum of Postgraduate Diploma programme in Dairy Management.
3. MANAGE (Centre on Management of Agricultural Extension), National Institute of Rural Development, Rajendranagar, Hyderabad in preparation of suitable curriculum for the Extension Management Course and Foundation Course on Extension Management.

3.1.3 Transfer of Technology Systems and Policies Unit

Background/Introduction

The primary function of this unit is to impart training and conduct research on Technology Transfer System. The case studies on existing methods of transfer of technology such as Operational Research Projects, Lab to Land, KVK System are taken up by the Unit.

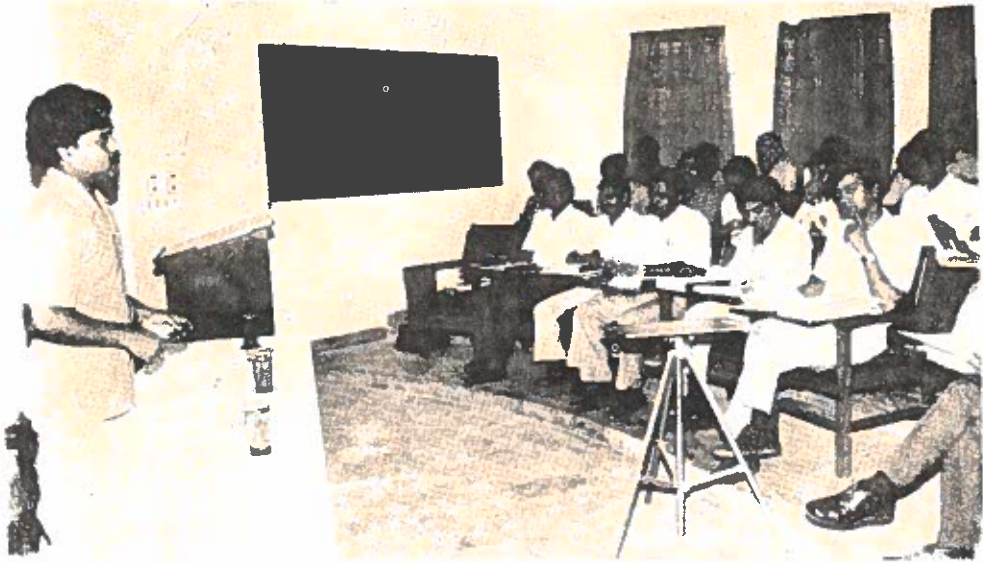
Objectives, Tasks and Functions

1. Training in Technology Transfer Processes and Appropriate Technology selection.
2. Develop effective Agricultural Communication techniques and mechanism.
3. Identify bottlenecks and managerial deficiencies of TOT Systems and evolving more efficient processes.
4. Consultancy and Advisory Services.

Activities during the year 1986

Study on Management of Lab to Land Programmes of ICAR

The Unit proposed and undertook a study on the management of Lab to Land Programmes of ICAR during the year 1986. For



A session of the Foundation Course in Progress



Participants at Photographic Lab.

For this purpose the information regarding Lab to Land Programmes of ICAR have been collected from 66 Lab to Land Centres out of 106 Centres from all over the country through structured schedule devised by the Unit.

(Dr. R.K. Samanta)

Course material developed

The following course material were developed and delivered to various Courses, Workshops, Summer Institutes conducted at NAARM.

1. Role of Audio-Visual Aids for Effective Communication in Teaching-Learning Processes
2. Technology Transfer and Appropriate Technology
3. Developmental Communication
4. Institutional Structure for Technology Transfer
5. Communication and Effective Presentation Techniques
6. Rural Development and Rural Leadership
7. Role Play.

3.1.4. Project Management Unit

Background/Introduction

Project management is designed to integrate complex efforts and deals with planning, scheduling, directing and controlling of Institute resources for a project that has been established for the completion of specific goals and objectives. The project management utilizes the "System Approach" to management through the use of personnel belonging to different functional disciplines assigned to a specific project. The main objective of project management is to make the most efficient and effective use of the resources of manpower, equipment, facilities, materials, money, information/technology so that the Institute objectives and goals can be achieved as per schedule, budget and the desired performance, taking into consideration the ever changing environmental input factor.

The integrated planning and control systems assure that the functional disciplines will understand their total responsibility

towards achieving project needs; assure that problems resulting from scheduling and allocation of critical resources are known before hand. The problems that can jeopardise successful project completion can be forecast, so that effective corrective action can be taken to prevent or resolve the problems.

The Project Management unit provides an individual research worker the necessary tools and techniques by which he can find proven, workable solutions to problem that can occur in a project management.

Objectives, Tasks and Functions

The objectives of Project Management Unit are:

1. Training on Agricultural Research Project Management Techniques.
2. Case studies on planning, monitoring, controlling and evaluation of Agricultural Research Projects.
3. Network Analysis - Flow chart and scheduling of the project activities and the networks of the projects.
4. Project budgeting and financial control.
5. Consultancy and advisory services.

Activities during the year 1986

1. Course on Agricultural Research Project Management (CARPM)

Two courses on Agricultural Research Project Management were organised from March 3-15, 1986 and from May 26 - June 14, 1986 for the benefit of 48 Senior Scientists of ICAR in the cadre of S3 and S4 and the Professors and Directors of Research, Agricultural Universities.

Course Content

The following course contents were covered:

- Project Planning - Importance of planning long-term and short-term project in the management process and systematic project planning.
- Project Selection - Methods and criteria used for project

selection so that one can undertake better research projects in agriculture.

- Monitoring and Evaluation - Network methods for time scheduling; control and monitoring; and resource smoothening of Research Projects.
- Organisation Structure - Principles of organisation and the importance of technology; and environment in organisational design.
- Group Dynamics - Characteristics and classification of groups; group decision making; and inter-group problems in organisations.
- Leadership - Role of leadership in management; leadership styles suitable to Indian conditions.
- Conflicts - Types of conflict that occur in organisations; and the methods used to resolve the conflicts.
- Organisational Communication - Meaning, concept and importance of organisational structure; communication behaviour; organisation structure's effect on communication; avoiding resentment; principles of effective listening; and barriers to listening.
- Mass Communication - Television in rural communication; steps in developing a TV programme; and rural broadcasting.

7. National Agricultural Research Project (NARP)

Eight case studies on some of the NARPs were undertaken so as to monitor and evaluate their performance. This involved collection of information through structured questionnaires, visits and personal discussions and analysis of the large volume of data collected. These case studies are to be presented at the Workshop next year and will serve as a discussion base for the Workshop and help to identify the constraints and difficulties in developing and implementing the project.

(Dr.S.N.Saha & Dr.K.Venkateswarlu)

8. New Project Management Techniques

New Project Management Techniques viz. Activity Slack Bar Chart (ASBC) and Activity Bar Chart (ABC) were developed.

Modified Gantt charts have also been developed for controlling the progress of various developmental projects. Thirty cases illustrating PERT have been prepared. Three cases illustrating CPM have also been prepared.

(Dr.K.Venkateswarlu)

Course material developed

The following course materials were developed and delivered by the Unit to various Courses, Summer Institutes and Workshops conducted during the year.

1. Research Programme Planning
2. The Research Project
3. Project Appraisal
4. Conflict Management
5. Management of Research Projects with special reference to monitoring and control.
6. Agricultural Production Potential and constraints
7. National Agricultural Research Project
8. Energy Management in Agriculture
9. Biomass and Energy
10. Solar and Wind Energy
11. Research Project Management
12. Conflict Management Style.
13. Creativity and Problem Solving
14. Project Management through Networks
15. Management of Research Stations.
16. Impact of Green Revolution and Strategies for future.

3.1.5. Human Resources Development Unit

Background/Introduction

People form the most important resource in any organisation. Human Resources Development unit is concerned with methods

and approaches for the growth and development of the people in organisations towards higher level of competency, creativity and fulfilment. The concept of development covers not only individuals but also groups. The human resources approach helps people grow in self control, responsibility and other abilities and then creates a climate in which all employees may contribute their best for improved results and efficiency. This is applicable to research organisations and scientists. Human resources development is attempted through training, feedback and counselling. Human resources are underutilised in many organisations.

Objectives, Tasks and Functions

1. To improve the research productivity and efficiency of Agricultural Scientists and Teachers through behavioural changes by training in human resources development and organisational behaviour. They are exposed to following topics:

Motivation
Interpersonal relationships team work
Leadership
Conflict Management etc.

2. To conduct research in the areas of Human Resources Management and Organisational Behaviour relevant to Agricultural Research and Education Management.

3. To prepare case studies, structured experiences for training in human behaviour, audio-visual aids and Role Play for effective training.

4. To provide consultancy in the areas of Human Resources Management and Organisational Behaviour.

Activity during the year 1986

Study on the Organisational Climate and Research Productivity in ICAR Institutes:

The Unit initiated a study on the organisational climate in ICAR Institutes. Three questionnaires were developed for this study separately, one for the Directors of the Institutes, one for Scientists and one for the technical staff. The questionnaires were sent to the Directors, 400 scientists and 300 Technical staff. By December, 1986 replies for questionnaires were received from 100 Scientists, 28 Technical Staff and one Director only.

(Dr.K.V.Subba Rao)

Course material developed

The following course materials were developed and delivered by the unit to the various courses, Summer Institutes and Workshop conducted during the year 1986.

1. Positive Thinking
2. ICAR History, Growth and present structure
3. Agricultural Research Service
4. Agricultural Research Systems of the World
5. Structure and design of organisations.
6. Individual and the organisation
7. Groups in organisations.
8. Group Dynamics
9. Leadership styles
10. Team Building
11. Fisheries Research and Development Programmes in India.
12. Motivation and Behaviour
13. Motivation and work
14. Motivation of Scientists
15. Interpersonal relationships
16. Rapid reading and comprehension
17. Organisational Behaviour
18. Making practicals learning oriented
19. Preparation of video lessons
20. Preparation of Audio Lessons.
21. Team-work in Research

22. Interpersonal Relationship

23. Transactional Analysis at work

24. Stereotypes.

4.1.6. Information and Documentation Unit

Background/Introduction

Information and documentation form the most critical inputs in any form of management activity. Such organised scientific information base is very important for decision making at all levels. Besides, a personal scientific information and retrieval system is necessary for every scientific worker to function as a more effective researcher.

Objectives, Tasks and Functions

1. To function as a repository of ideas and information, both national and international, in the field of Agricultural Research Management and Education and develop as a clearing house for dissemination and utilisation of such information to the clientele institutions.
2. Training Scientists in the personal scientific information management including the user education users programmes.
3. To develop appropriate management information systems for scientific management.
4. To coordinate for the sectoral information centres in India in the field of Agricultural Sciences.
5. To bring out learner's information packages and transferring Workshop/Training kits.
6. Agricultural Research Management Abstracts.
7. Consultancy and Advisory Services.

Activities during the year 1986

Technical Services

The Information and Documentation Unit attempted to collect important and useful literature and documents on Agricultural

Research Management, Planning, Development, Technology Systems and Policies, Project Management; Educational Technology; Information Management and other related areas. At present the Unit has about 15,000 publications and 400 Journals on Subscription/Exchange basis.

The Unit has many unique features in its information storage and retrieval. The following specialised corners have been formed for the effective utilisation of the documents:

Foreign Tour Reports Corner

In order to have an access for the Information on Agricultural Systems (Studies of the different countries, a specialised corner for Foreign Tour Reports of the ICAR/DARE Scientists has been established. A copy each of the tour reports, after submission to ICAR, is being sent to NAARM and these are organised countrywise with additional entries on card form by Scientist's name, Institutes and areawise. It is planned to introduce reprographic services for such information and also to microfilm these reports for effective storage and utilisation.

Repository Centre

Efforts have been made to establish repository centres of all institutes coming under CGIAR group as also many international organisations engaged in Agricultural Research, Development, Training, Education etc. This corner has preserved all the important publications of the International Agricultural Research Institutes for the benefit of all engaged in Agricultural Research and Development. Periodically catalogues of these holdings will be made available for effective utilisation.

Reprography

The Unit has three sophisticated plain paper copiers with enlargement and reduction facilities and three electronic stencil cutters. These reprographic machines meet the entire demand of the Academy and the users. About 1,70,000 copies were reproduced from the plain paper copiers and 1,000 electronic scanning (Stencil) were cut for the use in training programmes.

Training

Scientific information management is an integral part in almost all the training programmes organised at the Academy.

The Unit has developed many interesting topics for the benefit of Scientists in the ICAR Institutes and the Faculty Members of the Agricultural Universities. Different programmes were developed based on the levels of participating scientists. The sessions dealt were problem oriented and emphasised practical use.

Agricultural Research Management Abstracts (ARMA)

This publication is for the benefit of those engaged in Agricultural Research Management, Administration, Education and Development. ARMA covers many interesting areas in Agricultural Research Management covered under the different units of the Academy.

The material for two issues of 1986 has been released and sent for publication. The ARMA is brought out by Information and Documentation Unit of the Academy and is compiled and edited by Sri R.B.Gaddagimath, Librarian of the Academy. About 700 copies are made and circulated on exchange basis.

Course material developed

The following course materials were developed and delivered to various Courses, Workshops and Summer Institute conducted at the Academy.

1. Writing of Scientific Paper.
2. Transfer of Scientific Information and tips for writing objective research papers.
3. Citations, Citation Indexing and Science Citation Index.
4. Management of Personal Documents and Uniterm Indexing.
5. Frontiers in Agricultural Information Services at National and International level.
6. Review of Paper Writing and Index to Scientific Reviews.
7. Proof Reading and Scientific editing.
8. Use and Evaluation of Secondary periodicals and reference sources in Agricultural Services.
9. Poster sessions as a medium of Research Communication and Effective Conference Planning.

10. Book Reviewing in the Sciences and tips for effective science book making.
11. Research Front Specialities in the Sciences and Index to Scientific Research.
12. Technical Report Writing.
13. Scientific Information Management Tools as an Aid for Curriculum Design and Development.
14. Information sources and Management in Extension studies.
15. Planning of user education in Home Science Institutes.
16. Information use of Agricultural Scientists.
17. SCI as a tool for information management.
18. Review literature and ISR.

Workshops

Workshop on Information Services Management in ICAR Institute Libraries and Information Centres

A four-day Workshop on Information Services Management in ICAR Institute Libraries and Information Centres was organised from July 8-11, 1986 to discuss the course of action for an improved information services for the agricultural scientists in India. Sh. R.B. Gaddagimath coordinated the Workshop.

The Workshop highlighted the improvement of the library services for the scientists and modernizing the libraries to meet the changing demand and needs. During the Workshop the participants discussed the systematic organisation of literature, technological developments in information handling techniques including computer application for storage and retrieval to provide effective information services to scientists. The following topics were covered during the Workshop: Information needs of the Agricultural Scientists; Management Information Systems in Agriculture; Planning and Organisation of In-house documentation services; Education and Training for Agricultural Information handling and Sectoral Information Centres in Agricultural Sciences.

Consultancy and Advisory Services

The Consultancy and Advisory Service was rendered by the Unit to a number of Institutions at the National level.

3.1.7. CENTRALISED SERVICES

(1) Unit of Farm Operations and Management and the Agro-technology Demonstration Blocks

Background/Introduction

The Academy is landscaping its campus as a model for research and educational institutions. This is done, conscious of the fact that a beautifully landscaped campus leads to a comfortable environment and provides for improvement in micro-climate. The psychological comfort of green and sylvan surroundings promotes an academic atmosphere and leads to higher productivity. In landscaping the campus, stress is being laid in utilizing a large and varied collection of trees, shrubs, climbers, herbaceous plants and ground covers with an eye on the educational value of a well labelled rich collection of flora.

The Agro-technology Demonstration blocks of NAARM have been conceived as the show window of the National Agricultural Research System. The latest technology and crop varieties being released by the different ICAR Institute and Agricultural Universities are being put on show here in one place (subject to agro-climatic limitations) for the benefit of visitors and trainees of NAARM. The visitors include National and International Scientists, Policy makers and Administrators of Agricultural Research and Development.

Objectives, Tasks and Functions

The Farm Operations and Management is a centralised service of the Academy and has the objectives of landscaping of the campus, developing and maintaining the estate and laying out the agro-technology demonstration blocks in different areas of agriculture, which includes horticulture, sylvi systems, and field crops. The unit undertakes all tasks and functions to achieve the above objectives. The landscaping activities cover the entire 50 ha. of the campus and the agro-technology demonstration laid out in 15 ha. of land.

Activities during the year 1986

Landscaping

The avenues on the major roads which have been planted are being maintained. These include tree species of about 15 genera. The landscaping of the administrative building has been completed. The garden development of the Academic block and the Halls of Residence is in progress. A rose garden of over a thousand bushes has been developed which has been adjudged as the best maintained by any Institution in the twin cities, by the Hyderabad Rose Society.

Horticultural Crops

(i) Fruit Crops

The fruit orchards of six different fruit crops have been established. All these are improved varieties released by ICAR Institutes/Agricultural Universities. The guava block is planted with the varieties of Allahabad Safaida, Lucknow 49, Safaid Jam, Kohir Safaida and Banarsi Surkha.

The Sapota plot has been established with the varieties of Oval, Cricket Ball, Bombay and Kalipatti.

The Grape orchard has been planted with Gulabi, Cardinal, Champion, Beauty Seedless, Perlette, Delight, Thompson seedless, Pusa seedless, Himrod, Banqui-abad, Early Muscat, Raisina, Kishmis Rosovis, Golden Queen, Arka Mans, Arka Shyam, Arkavati, Arka Kanchan, Dilkhus and Anab-e-shahi.

The Ber plot has been planted with the varieties Gola, Seb, Mundia, Kaithaly and Umran.

The Pomegranate varieties grown are Jodhpur red, Basin seedless, Jyoti (G.K.V.K.-1) and Ganesh.

The Citrus cultivars planted include Moosambi, Santara, Kagazilime, Khasi Mandarin, Sathgudi etc.

(ii) Vegetable Crops

Twenty two varieties in eight kinds of vegetables and forty three varieties in sixteen kinds of vegetables have been grown in the Khareef and Rabi season of 1986 respectively. The different kinds of vegetables grown were tomato, brinjal, chillies, bhendi,

cucumber, bottle gourd, bitter gourd, sponge gourd, ridge gourd, french beans, peas, cabbage, cauliflower, knolkhol, radish and carrot.

(iii) Floricultural Crops

Half a hectare block has been prepared for the demonstration of the commercial cultivation of floricultural crops. The crops include Rose, Jasmine, tube rose and bulbous plants.

Sylvi Systems

Besides the conventional input systems for increased productivity, different systems of plantations have been evolved to increase the productivity efficiency of land and obtain food, fuel and fodder from the same piece of land. NAARM has established the following demonstrations for the benefit of the course participants from different parts of the country. Various sylvi systems that have been established are as follows:

(i) Agro-forestry system

It is a purely agricultural system in which tree component is introduced in appropriate spacing so that the available space can be utilized for growing food crops based on the principles of mutual benefit in terms of input utilization and association.

This consists of an area of 200 m x 50 m with alternate rows of Leucaena leucocephala (Subabool) (K 8 variety) and Acacia auriculiformis planted at a spacing of 5 m x 2 m. Both these species are quick growing and drought tolerant. Cowpea variety C-86 was grown in the space between the rows.

(ii) Agro-horticultural system

This comprises of 100 m x 50 m plot planted with the Balanagar variety of Custard apple at the spacing of 5 m x 2 m and the area between the rows are being used for growing field crops like green gram (Var P-16) and black gram (T-9).

This system will supply parallel production of fruit and field crops and added income to the farmer.

(iii) Sylvi-pastoral system

This system has been designed to produce fodder for stall feeding/grazing purposes and supply of timber and fuel. An area

of 180 m x 50 m unsuitable for cultivation has been planted with Azadiracta indica (Neem) plants at 5 m x 5 m and the complete area is sown with Stylosanthus hamata which is leguminous quick growing and drought tolerant fodder crop.

This system will demonstrate the effective utilization of most unproductive land for economic benefit of farming community.

(iv) Horti-pastoral system

It is a flexible system in which unfertile and unproductive land can be utilized for production of fruit crops and developed as a pasture either for grazing or stall feeding purposes. In this system Emblica officinalis and Phalsa will be planted with the space between the rows sown with Stylosanthus hamata and Cenchrus ciliaris in alternate patches. After about four years all these fruit crops will start giving fruits and pasture also will be established for its full utilization.

(v) Energy Plantation

An area of 100 m x 50 m has been planted with a recommended variety of Leucaena leucocephala (Subabool) at 2 m x 2 m spacing. This highly density plantation will be established to demonstrate the production of fodder and fuel in rural areas. This will encourage the small and marginal farmers to raise cattle and other animals for meat, milk and drought purpose with the resources produced at their own farm.

Field Crops

(i) Oilseeds

The demonstration laid out in Khareef, 1986 in oilseeds include groundnut and sunflower. Two varieties of groundnut i.e. T.M.V.-2 and K-3 were sown and in sunflower also two varieties viz. E.C.68414 and Morden were demonstrated.

(ii) Millets

Three varieties of two millets were grown. In Jowar the varieties grown were C.S.H. 9 and S.P.V. 346 and in Bajra, the variety W.C. 75 was grown.

(iii) Cereals

Two varieties of Maize viz. Garga-5 and D.H.M.103 were grown.

(iv) Pulses

Two varieties of Redgram L.R.G.30 and C-11 were cultivated.

Non-conventional Energy Systems

(i) Domestic bio-gas plant

A domestic bio-gas plant developed as a semiurban unit by Dr.K.G.Gollakota (formerly of the A.P.Agricultural University) is under continuous demonstration at the Academy. In this unit, non-edible oil cakes are used to produce the combustible gas which is fed into a store and is used for everyday use. (Corder 1983)

(ii) Wind Mill

A wind mill has been installed in the campus to harvest the wind energy for running a water pump.

(iii) Solar Water Heater

A multi-flat collector water heating system has been installed in NAARM Hostel for supplying hot water at 65°C. This hot water is being used for cooking and cleaning utensils.

Consultancy and Advisory Services

The unit has been providing consultancy and advisory services in landscaping and ornamental horticulture to the nearby government institutions such as S.V.P. National Police Academy and Indian Grain Storage Research Institute.

(II) Production and Publications

Academy has taken up several publications on its own to help the research managers in keeping themselves in touch with the latest techniques of management. The Academy has got an offset press used for publishing the materials developed by the Faculty for training purpose. Some of these publications are mentioned below:

1. Course materials and Reading materials for each course conducted by the Academy.
2. Case materials prepared on the basis of research studies undertaken by the Faculty of NAARM.

Green and Glory:

Green and Glory is a laboratory journal published for every Foundation Course based on the thesis of the trainees of the particular batch. It provides them an opportunity in writing of scientific and popular articles. For every issue 400 copies of the journal are printed and the copies are sent to all ICAR Institutes, Agricultural Universities and user agencies like various development departments of agriculture, animal sciences, dairy corporation and fisheries departments of the various states.

Sri V.E.Sabarathnam is the Chief-Editor of the journal. Green and Glory for the XXII batch of Foundation Course on Agricultural Research Management (FCARM) was published in November, 1986.

Newsletters

NAARM News: 'NAARM News' brings out the news and information about the events, programmes, future programmes and the welfare activities of the staff of the Academy and is published every quarter. Dr. Devesh Kishore is the Editor-in-Chief of this newsletter.

NAARM Alumni News: 'NAARM Alumni News' is a newsletter which publishes the information about the Alumni half yearly. Dr.R.K.Samanta is the Chief Editor of the newsletter.

(III) Administration and Accounts

Background/Introduction

For providing effective administrative services and in accounts matter, this unit has been arranged in six sections viz.

1. Admn.I Section - dealing with establishing matters.
2. Admn.II Section - dealing with Cash and Bills.

3. Admn.III Section - dealing with Stores and Purchase, Transportation and Budget matters.

4. Admn.IV Section - dealing with Diary and Despatch.

5. Admn.V Section - dealing with all training matters and compilation and publication of rules and regulations relating to scientific, technical, administration etc.

6. Audit & Accounts Section - dealing with all financial matters.

Objectives, Tasks and Functions

The Unit provides administrative and financial support for the smooth conducting of various training programmes of the Academy in addition to attending to the day-to-day administration of the Academy and the NAARM Faculty.

Activities during the year 1986

The necessary administrative financial support was provided for the conducting of the four Foundation Courses on Agricultural Research Management, two Orientation Courses on Agricultural Research Management and a number of Workshops held during the year. The Unit also rendered necessary help in the administration of the Academy.

Course material developed

The following course materials were developed for the various training courses conducted by the Academy.

1. Office Communication
2. Scientific Personal Policies
3. An overview of general conditions and benefits.

3.2. STAFF POSITION

Sl. No.	Category	Sanctioned strength upto March, 1985			Additions proposed in VII Plan as recommended by IWSU
		Plan	Non-plan	Total	
1.	Scientific	16	05	21	22
2.	Technical	12	03	15	38
3.	Administrative	17	12	29	24
4.	Auxiliary	05	04	09	13
5.	Supporting	10	07	17	41
Total		60	31	91	138

3.3. FINANCIAL OUTLAY:

(Rupees in lakhs)					
Sl. No.	Head	VI Plan actuals		VII Plan requirement	
		Plan	Non-Plan	Plan	Non-Plan
1.	Pay & Allowances	16.81	21.50	125.10	117.52
2.	T.A.	1.09	0.90	10.00	10.00
3.	Contingency	14.45	44.54	25.80	111.75
4.	Equipment	32.62	--	62.50	--
5.	Works	212.67	--	277.10	--
6.	Vehicles	4.55	--	7.00	--
7.	Library Books	11.94	--	12.50	--
8.	Others	--	1.61	--	10.73
Total		294.13	68.55	520.00	250.00

4. UNDP Advanced Centre on Agricultural Education and Research Management

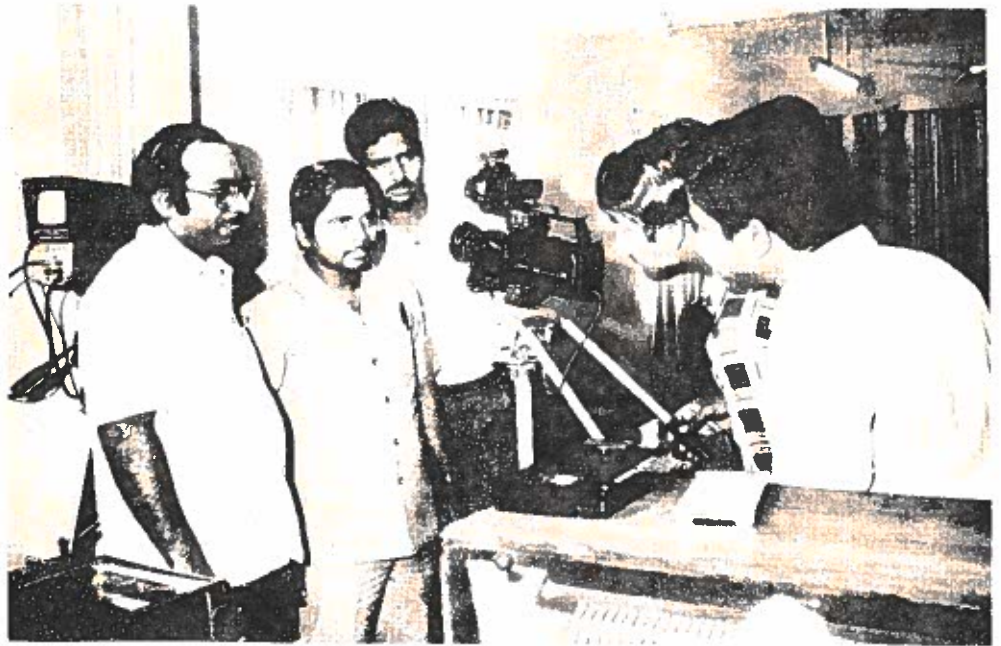
An Advanced Centre on Agricultural Education and Research Management has been established in 1986 at the Academy with assistance from UNDP/UNESCO/FAO with the primary objective of establishing a system for managing agricultural education and research so as to ensure continuous adjustments to changing national needs through scientific study of educational research management process and training key personnel in their application. The focus of the Centre will be to develop and manage agricultural research and education to meet the national needs which takes in to account the capabilities of farmers and addresses the needs for forecasting research trends adjusting research policies leading to more effective education of agricultural researchers and educators by developing their managerial skill. This will also include a continuous liaison and consultancy with ICAR Institutions and Agricultural Universities.

In order to make research more meaningful, mission and theme-oriented, and to ensure that expenditure on research is effectively utilized, training in Project Management is required. This includes exposure and training not only on Project identification selection, formulation, monitoring and evaluation, but also on interdisciplinary approaches, group dynamics, interpersonnel relations and conflict management. This Advanced Centre will provide training in these areas, as part of the institutional objectives and mandate.

In agricultural education, which is field oriented, learning and doing should go hand in hand, and the learning and teaching process should be systematized through modern methods of pedagogy. Evaluation of a proper instructional technology for various disciplines in agriculture requires intensive survey and planning and indepth studies on the existing methodologies to evolve better methods of teaching. The Advanced Centre would fill this vital gap.

An effective interaction between the researchers and farmers would ensure a greater awareness of the constraints and problems of farmers in the design of suitable technologies. Such a close linkage is possible by adopting the Farming Systems Research (FSR) approach, under which the multi-disciplinary team of scientists and farmers would be equal partners in identifying technological alternatives more relevant to the farmers environment. Exposure of the agricultural research scientists in the country to FSR would make their programmes more meaningful and purposeful. This task could be undertaken by the Advanced Centre through training based on case studies and field demonstrations.

The Advanced Centre on Agricultural Education and Research Management while providing in Agricultural Research Management and Educational Technology will fulfil a long felt need for developing expertise, since such programmes do not exist elsewhere in the country.



Participants at CCTV Lab.



Library Consultation by the Trainees

5. PHYSICAL FACILITIES

The Academy has set up the following laboratories to make the teaching-learning process more effective and help the trainees in effective use of mass media for dissemination of information.

5.1. Closed Circuit Television System Lab

It has been established to provide intensive training to senior scientists in preparation and to conduct role play exercises for personality development. The laboratory is also used for sharpening individual communication skills and to improve the group communication process in building up an efficient research team. Educational programmes related to structured learning are also being produced.

5.2. Audio-visual laboratory including Photography Unit

This provides an excellent opportunity for scientists to have practical experience in use of A.V. Aids for effective presentation of scientific material during seminars, talks, conferences etc. Scientists learn about preparation of various types of A.V. materials such as transparencies, slides etc.

5.3. Mass Communication Laboratory

The scientists are being trained in preparation of material for oral communication. They prepare scripts for broadcasting purpose. They record their programmes such as interviews, discussions and talks specially from the rural listener's point of view and devise ways and means of improving the effectiveness of the programmes.

5.4. Organisational behaviour laboratory

The Academy has an organisational behaviour laboratory for exercises in various aspects of organisational behaviour.

5.5. Computer Facilities

The Academy has acquired UPTRON S-850 micro-computer. It has softwares such as BASIC, FORTRAN and COBOL languages, Assembler, Graphics, Word Processor, Information Processing and Data Processing. This computer has been installed to develop knowledge and skills of the trainees in one of the peripheral areas of training on computer use. The computer is also to be utilised for preparing NAARM Alumni Directory, Information storage and retrieval on Agricultural Research Management and data analysis of the research projects undertaken by the Faculty.

5.6. Bank Counter

An extension counter of the Central Bank of India, Golconda Fort Branch, Hyderabad started functioning at NAARM in the year 1986 to cater to the banking requirements of the Academy Faculty Members, Staff and Trainees.

6. NAARM ALUMNI ASSOCIATION

The Quinquennial Review Team of NAARM had recommended that an Alumni Association may be organised.

Consequent on this recommendation, a NAARM Alumni Association, was formed on 15th June, 1984 and subsequently registered under A.P. Public Societies Registration Act vide 127 of 1985.

The main objectives of the association are:

- (i) to develop a feeling of fraternity among all the NAARM Alumni and to provide a forum for social get together among themselves.
- (ii) to encourage scientific activities that symbolise the excellence in Agricultural Research Management and allied disciplines.
- (iii) to disseminate the latest information in Agricultural Research Management to enhance the professional skills of scientists, administrators, educators and research managers by organising conferences, seminars, workshops, symposia etc.

About 350 scientists of ICAR and Professors of Agricultural Universities have enrolled their names as members of the association.

The office bearers of the Association for the year 1986-88.

Honorary Patron:	Director General, ICAR
President:	Dr.K.V.Raman, Director, NAARM
Vice President: (Hyderabad Region)	Dr.M.Rajagopalan, Faculty Member, NAARM
Vice President: (General)	Dr.R.D.Rawal, IIHR, Bangalore
General Secretary & Editors-in-chief:	Dr.R.K.Samanta, Faculty Member, NAARM
Joint Secretaries:	
South Zone:	Sh. Pradyumn Kumar, IIHR, Bangalore
Central Zone:	Sh. R.S.Khatri, IASRI, New Delhi.

North Zone: Dr.Rishendra Verma, IVRI, Izatnagar.

East Zone: Dr.Padhi, CRRI, Cuttack.

NAARM Alumni Association proposes a National Seminar on Agricultural Research Systems and Management in the 21st Century to be held at NAARM from December 8-10, 1987.

7. PUBLICATIONS

7.1 Scientific papers published

Balaqura, T. and Rajagopalan, M., 1986 Management of Agricultural Research Project in India 1 : Project Planning. Agricultural Administration 22:217-231.

Balaqura, T. and Rajagopalan, M., 1986 Management of Agricultural Research Project in India: 2 Research Productivity, Reporting and Communication. Agricultural Administration. 23:1-15.

Dwivedi, R.S., Joshi, Y.C., Saha S.N. Thakkar, A.N. and Kordia, V.G., 1986 Canopy orientation and energy harvest in groundnut (Arachis hypogaea L.) under water stress Oleagineux 41(10):451-455.

Gaddagimath, R.B. 1986 Selection of Scientific Journals based on Information use. A case study of Sugarcane Crop. Indian J. Agric. Library Information Sci. 1984.

Samanta, R.K., 1986 Problems of Transfer of Agricultural Technology. Vision 4(5):1-11.

Samanta, R.K., 1986 Preparation of work environment of Agricultural Research Service Scientists in ICAR Institutes. Indian J. Extension Education. 21:53-61.

Samanta, R.K., 1986 Effective Agricultural Communication Strategy Need for Efficient Transfer of Agricultural Technology. Journal of Communication (In Press).

Samanta, R.K., 1986 Scientists' perception of motivational climate in Agricultural Research organisations vis-a-vis their satisfaction and role perception : The Indian Scene. Journal of Management Development Studies (In Press).

7.2 Books, Bulletins, Monographs, Technical Reports and Notes

i) Information Services Management in ICAR Institutes, Libraries and Information Centres (Theme papers). July 8-11, compiled and edited by Sh.R.B.Gaddagimath.

ii) Background readings in Agricultural Information Services Management. Compiled and edited by Sh.R.B.Gaddagimath.

7.3 Papers presented at National and International Symposium/Seminar/Conference/Workshops.

- * Gaddagimath, R.B. 1986. Information Services for Agro-forestry research and issues for organising Agro-forestry research in India (workshop paper) ICAR/ICRAF Training-cum-Workshop on Agro-forestry, Central Research Institute for Dryland Agriculture, Hyderabad, September 16-30.
- * Raman, K.V. 1986. Designing appropriate curricula and courses: constraints and considerations.
- * Raman, K.V. 1986. Management of Human Resources in Agricultural Research. Indian Efforts and Experiences. International Workshop on Managing Human Resources in Agricultural Research held at Dhaka, March 3-5.
- * Raman, K.V. 1986. Productivity in Agriculture, at ASCI, Hyderabad, March 11-12.
- * Raman, K.V. 1986. Surface chemistry of soil minerals with respect to their reactivity. Prof. J.N.Mukherjee - ISSS Foundation lecture. TNAU,Coimbatore, September 23.
- * Raman, K.V. 1986. National Agricultural Research System in India and its impact on Agricultural Production and productivity. First International conference on National Agricultural Research System and Second Global convention of IFARD, Brasilia, Brazil, October 6-11.
- * Raman, K.V. 1986. Trends in Agricultural Research and Extension in India, Kuala Lumpur, Malaysia, November 27-30.
- * Raman, K.V. and Sabarathnam, V.E. 1986. Training in Agricultural Research Management for Development of Research Personnel, presented at the National Seminar on Extension Education and Human Resource Development organised by the Indian Society of Extension Education at APAU, Hyderabad. November 14-17.
- * Samanta, R.K. 1986. Effective Agricultural communication strategy - Need for Efficient Transfer of Agricultural Technology, Seminar on Agricultural Communication Research held at G.B.Pant University of Agriculture & Technology, Pantnagar, January.

7.4. Special Lectures Delivered

Dr. Devesh Kishore

- * 'Mass Media in Rural Communication' in the orientation programme on cooperative philosophy and Rural Communication, organised

- forestry
in India
1. Agro-
2. Hyde-
- courses:
- icultural
Workshop
held at
- derabad,
- respect
lecture.
- in India
ty. First
System
October
- Extension
- icultural
ersonnel,
d Human
Extension
- strategy
Seminar
iversity
- rogramme
organised
- by Vankunt Mehta National Institute of Cooperative Management,
Poona.
- 'Organisational Communication in Extension' in the Course entitled Management of Agricultural Extension organised and conducted by National Institute of Rural Development, Hyderabad.
 - 'Curriculum Design and Development for Higher Education' at the Annual Conference of Indian Association for Educational Technology held at New Delhi.
 - 'Principle of Effective Communication - Listening' in the Course on Training of Trainers at SVP National Police Academy, Hyderabad.
 - 'Communication and Motivation for Rural Development' in the Course on Integrated Rural Development conducted by State Institute of Rural Development, Hyderabad.
 - 'Experimental Design on Media Research' at the Development Communication Seminar organised by University of Poona.
 - 'Communication Process for Rural change in the Course on Training Methods and Techniques for Rural Development conducted by the National Institute of Rural Development, Hyderabad.
 - 'Communication in Training' in the Training Programme on Training Methods and Techniques in Rural Development conducted by the National Institute of Rural Development, Hyderabad.
 - 'Developing Effective Communication Skills in Trainees' in the Course designed for training administrators by SVP National Police Academy, Hyderabad.
 - 'Organisation Communication' in the Course on Extension Management conducted by Directorate of Extension, University of Agricultural Science, Bangalore.
 - 'Agricultural Research Management Process' in International Course conducted by ICRISAT, Hyderabad.
 - 'Principles of Educational Technology in developing/modifying a course' at Summer Institute on Dairy Business Management, conducted by NDRI, Karnal.
 - 'Role of Communication in Transfer of Technology' in the Course conducted by Directorate of Rice Research, Hyderabad.

- * 'Communication Methodology' at the Training Programme for Indian Forest Service Officers organised and conducted by NIRD, Hyderabad.
- * 'Innovative Training Methodology' in the Course entitled Training Methodology conducted by Extension Education Institute, A.P.A.U. Hyderabad.

Dr.K.V.Subba Rao

- * 'Groups in Organisations' to subject matter specialists in a programme conducted by Extension Education Institute of APAU, Hyderabad.

Dr.S.N.Saha

- * 'Agricultural Production Potential and constraints' at State Institute of Rural Development, Hyderabad.
- * 'Project Management' in International Course conducted by ICRISA Hyderabad.

Dr.R.K.Samanta

- * 'Technology, Transfer and Rural Development and its Philosophy' in the State level training programme of subject matter specialists of various states on Rice Production Technology conducted by Directorate of Rice Research, Hyderabad.

Sh.V.E.Sabarathnam

- * 'Integrated communication approach for agricultural development' for the course on Agricultural development for Agricultural Officers at State Institute of Rural Development, Rajendranagar, Hyderabad.

Dr.T.Balaguru

- * 'Green Revolution Game' in the Course on Management of Agricultural Extension for Master Trainers conducted by National Institute of Rural Development, Hyderabad.

Dr.M.M.Anwer

- * 'Landscaping and beautification of the campus' at SVP National Police Academy, Hyderabad.

for Indian
Hyderabad.

Sh. B. B. Gaddagimath

1 Training
A.P.A.U.,

- 'Organising information centres for training development in Police Training Organisation' at SVP National Police Academy, Hyderabad.

- 'Information service management for Extension Research' at National Institute of Rural Development, Hyderabad.

in a pro-
AU, Hyde-

3 Institute

ICRISAT,

'philosophy'
specialists
by Direc-

velopment'
il Officers
Hyderabad.

f Agricul-
l Institute

' National

8. MISCELLANEOUS

8.1. Faculty Development

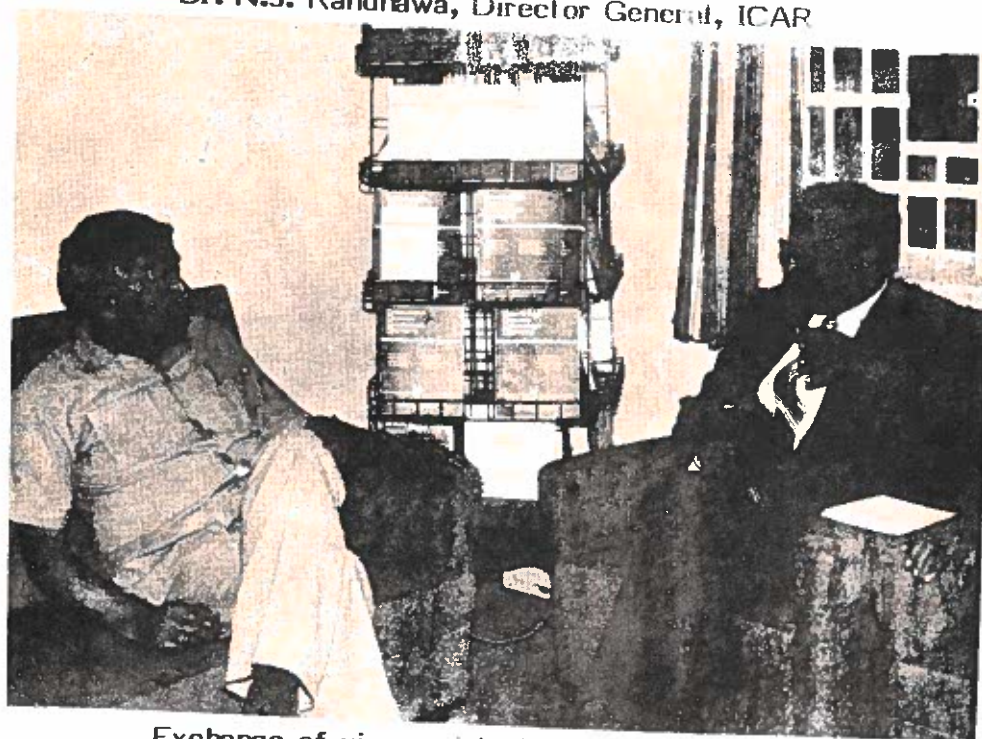
- * Dr.M.M.Anwer, Associate Professor attended a Training Programme on 'Management of Agricultural Research' conducted by the United States Department of Agriculture at Washington D.C. from July 28 - September 5, 1986. The course aimed at development of knowledge and skills in Agricultural Research Management.
- * Sh.K.Prathap Reddy, Associate Professor attended the "Fellow Programme in Management" at Indian Institute of Management Ahmedabad.
- * Dr.S.N.Saha, Professor participated in training programme on Energy Management in Agriculture organised by India Development House New Delhi.

8.2. Foreign visits

- * Dr.K.V.Raman, Director, NAARM participated in the 'International Workshop on Management of Human Resources in Agricultural Research' at Dhaka, Bangladesh from March 3-5, 1986 and presented the country paper on India entitled 'Management of Human Resources in Agricultural Research - Indian Efforts and Experiences'.
- * Dr.K.V.Raman, Director, NAARM participated in the XIII International Congress of Soil Science held at Hamburg, Federal Republic of Germany from August 13-20, 1986. Also Dr. Raman visited International Service for National Agricultural Research in Netherlands and discussed with Dr.Alexander Von Der Osten, Director General, ISNAR regarding collaborative programmes in Agricultural Research Management.
- * Dr.M.M.Anwer, Associate Professor attended a six weeks programme on Management of Agricultural Research from July to September 5, 1986 conducted by the U.S. Department of Agriculture at Washington D.C.
- * Dr.K.V.Raman, Director, NAARM attended the First International meeting of the National Agricultural Research Systems and Second IFARD Global convention in Brasilia, Brazil from October 5-11, 1986 and presented the country paper on India entitled 'National Agricultural Research System in India and its impact on Agricultural Production and Productivity'.



Dr. K.V. Raman, Director, NAARM explaining the salient activities of the Academy to Dr. N.S. Randhawa, Director General, ICAR



Exchange of views with the foreign dignitaries

- * Dr.K.V.Raman, Director, NAARM led a four-member delegation to the Regional Seminar organised by Commonwealth Association of Scientific Agricultural Societies and Agricultural Institutes of Malaysia at Kuala Lumpur from November 27-30, 1986 and presented the country paper entitled 'Trends in Agricultural Research and Extension in India'.

8.3. Distinguished Visitors

- * Alois C. Mtowa, Senior Consultant in Project Management, Eastern and Southern African Management Institute, Arusha Tanzania.
- * Dr.Tafar Altaf, Secretary, Agriculture, Government of Punjab, Lahore, Pakistan.
- * Mr.Mohammed Nabisher Zai, President, Design and Coordination Department, State Planning Committee, Kabul, Afghanistan.
- * Mr.Homayun Fierozan and B.R.Chaudhury, Joint Secretary (O&M), Government of Bangladesh.
- * Mr.B.Bhattaraj, Under Secretary, Ministry of Finance, HMEN/ Kathmandu, Nepal.
- * Mr.William M. Rivera, FAO/University of Maryland.
- * Mr.Billy L. Coffindaffer, FAO/University of Maryland.
- * Mr.A.Venkataraman, Secretary & Commissioner, Agriculture, Madras, Tamilnadu.
- * Dr.Shirley White, World Bank/FAO consultant, MANAGE, Cornell University, Ithaca.
- * Dr.Ester Zulberti, Training Officer, International Council for Research in Agro Forestry (ICRAF) Nairobi, Kenya.
- * Mr. Bandhopadhyaya, Secretary, Department of Rural Development, Ministry of Agriculture and Rural Development.
- * Mr. B.N.Rahalkar, Land Scape Architect from Bombay.
- * Dr.A.Mukhopadhyay, S-3, IIHR, Bangalore.
- * A five member High Level Delegation of Agricultural Scientists from Pakistan headed by Dr.Amir Muhammed, Chairman, Pakistan Agricultural Research Council.

- * Mr. Lawrence Busch, Department of Sociology, University of Kentucky, Washington USA.
- * A delegation from Ethiopia comprising Yoseph Woldemichel, Petros Wontamid Binu Abede, Besrat Bemissie Makon nin mucat, Abefaw Assefa, Berhanu Debele and Solomon Bellele.
- * Dr.Kohotyanovilch, USSR, Inst. Agric. Microhol Leningrad, Pushkin-6.
- * A four members Delegation of Agricultural Scientists belonging to China headed by Mr.Sun Xiang, Department of Education, Ministry of Agriculture, Animal Husbandry and Fisheries (MAAF).
- * Mr.Lu Xunhao, Beijing Agricultural University.
- * Dr.Richard Blue, Director USAID, India.
- * Dr., Ron Pollock, USAID, New Delhi.
- * Dr.Guy Baird, Winrock International.
- * Dr. Floyd Williams.
- * Dr.Chuck Antbold.
- * Kathy Allison, USAID, OICD.

8.4. Honours and Awards

- * In Rose show organised by Hyderabad Rose Society, NAARM bagged five first, two second and three third prizes.
- * The Tophies for 'Best Rose Garden' maintained by an Institution and the Best display of Flowers was won by the Academy.
- * In a show organised by the All India Industrial Exhibition Society, NAARM bagged 14 first and seven second prizes. These prizes were awarded for the fruits, vegetables, cutflowers and pot plants exhibited by the Academy in the different classes of the show.
- * In the Horticultural show organised by the Golconda Society of Hyderabad at the Qutub Shahi Tombs, the Academy won four first and five second prizes. The materials exhibited were fruits, vegetables, cutflowers and potted plants.



Awards bagged by NAARM at the
A.I.I.E. Horticultural Show, 1986



Dr. K.V. Raman, Director, NAARM in conversation
with Shri Chandulal Chandrakar, Ex-Union State
Minister for Rural Development during his visit
to the Academy

8.5. Important Committees of the Academy

8.5.1. Management Committee

The Academy is guided in planning, organising and conducting of various types of training programmes and research studies by the Management Committee of NAARM. The Management Committee approves the programmes to be undertaken. The Academy ensures their implementation both in terms of physical targets and schedules and provides broad guidelines for future programmes. The Management Committee also helps in formulation and consideration of proposals for Five Year Plan and Annual Plan and takes periodical review of progress of development schemes. The composition of the Management Committee of NAARM is as follows:

- | | |
|---|----------|
| 1. Dr.K.V.Raman
Director, NAARM | Chairman |
| 2. Dr.Baidyanath Misra
Deputy Chairman
State Planning Board
Orissa Secretariat
Bhubaneswar. | Member |
| 3. Dr.H.R.Kalia
Ex.Vice-Chancellor
HPKV, Palampur. | Member |
| 4. Dr. B.Panda
Director
Central Avian Res. Instt.
Izatnagar. | Member |
| 5. Dr.T.P.Ojha
Director Central Instt. of Agric. Engg.
Bhopal. | Member |
| 6. Dr.R.S.Paroda
Director
National Bureau of Plant Genetic
Resources, New Delhi. | Member |
| 7. Dr.I.P.Abrol
Dy.Director General (SAE)
Indian Council of Agric. Res.
New Delhi. | Member |

Central instt. of Agric. Engg.

- | | |
|--|------------------|
| 8. Dr.Hanumantha Rao
Director
Animal Husbandry
Govt. of A.P.
Hyderabad. | Member |
| 9. Director of Agriculture
Govt. of Punjab
Chandigarh. | Member |
| 10. Dr.A.Appa Rao
Vice-Chancellor
A.P.Agric. University
Hyderabad. | Member |
| 11. Dr.Maharaj Singh
Dy.Director General (Educ.)
Indian Council of Agric. Res.
New Delhi. | Member |
| 12. Sri S.W.Oak
Director (Finance)
Indian Council of Agric. Res.
New Delhi. | Member |
| 13. Sri G.C.Sharma
Sr.Adm.Officer
NAARM, Hyderabad. | Member Secretary |

The Management Committee, besides assisting Director of the Academy in the discharge of his functions, particularly concentrates his attention on the Research and other programmes of the Academy and ensures their implementation both in terms of physical targets and time schedules, while examining these programmes periodically, pinpoints bottlenecks, if any, and suggests suitable remedial measures.

The tenth meeting of the Management Committee of NAARM was held on 25th July, 1986.

8.5.2. Steering Committee on Advanced Centre on Agricultural Education and Research Management

This Committee was initially called the Sub-project Advisory Committee and has been later renamed as Steering Committee.

The Committee will periodically review the progress and plan of the project and advise the project staff. This committee will meet at least once in a year and while reviewing the progress of the projects, will provide broad guidelines for its implementation based on the suggestions of user agencies. The following are the members of the Steering Committee.

- | | |
|---|------------------|
| 1. Dr.K.V.Raman, Director,NAARM | Chairman |
| 2. Dr.Maharaj Singh, DDG (E), ICAR | Member |
| 3. Dr.N.K.Ananta Rao,
Former Director, NAARM | Member |
| 4. Dr.Guy Baird, Winrock International | Member |
| 5. Dr.A.Appa Rao, VC, APAU | Member |
| 6. Dr.A.M.Michael, Director, IARI | Member |
| 7. Dr.R.Nagarcenkar,Director, NDRI | Member |
| 8. Dr.A.Rammohan Rao,
Dean PG Studies, APAU | Member |
| 9. Dr.M.Rajagopalan, Professor | Member |
| 10. Dr.Devesh Kishore, Professor | Member |
| 11. Dr.V.Raghavendra Rao, Professor | Member |
| 12. Sri G.C.Sharma, SAO | Member Secretary |

9. PERSONNEL

Dr.K.V.Raman

Director

Director Personal Cell

- | | |
|---------------------|-----------------|
| 1. Sh. P.P.Krishnan | Sr.Stenographer |
| 2. Sh. T.Sreenivas | Jr.Clerk |
| 3. Sh. P.Swamy | SSG I |

I. Agricultural Research Systems and Policies Unit

- | | |
|-----------------------|---------------------------|
| 1. Dr.M.Rajagopalan | Professor, Unit-in-charge |
| 2. Sh.V.E.Sabarathnam | Assoc. Professor |
| 3. Dr.T.Balaguru | Assoc. Professor |
| 4. Sh.J.S.R.Murthy | Stenographer |

II. Educational Systems and Technology Unit

- | | |
|-------------------------|----------------------------------|
| 1. Dr. Devesh Kishore | Professor, Unit-in-charge |
| 2. Dr.M.P.Singh | Professor (Joined on 17.12.1986) |
| 3. Sh.N.R.Nageswara Rao | A.V.Operator |
| 4. Sh. K.S.Sharma | Stenographer |

III. Transfer of Technology Systems & Policies Unit

- | | |
|---------------------------|----------------------------------|
| 1. Dr.R.K.Samanta | Assoc. Professor, Unit-in-charge |
| 2. Sh.L.Venkateswarlu | Photographer |
| 3. Sh.R.Satyanarayana Rao | Media Operator |
| 4. Sh.P.J.Pramod Kumar | Jr.Stenographer |

IV. Project Management Unit

- | | |
|-------------------------|-----------------------------------|
| 1. Dr.V.Raghavendra Rao | Professor, Unit-in-charge |
| 2. Dr.S.N.Saha | Professor |
| 3. Dr.K.Venkateswarlu | Professor |
| 4. Dr.P.Manikandan | Assoc. Prof. (Joined on 1.9.1986) |
| 5. Smt.L.Jhansi Lakshmi | Stenographer |

V. Human Resources Development Unit

- | | |
|-----------------------|---------------------------|
| 1. Dr.K.V.Subba Rao | Professor, Unit-in-charge |
| 2. Dr.M.M.Anwer | Assoc. Professor |
| 3. Mr.K.Prathap Reddy | Assoc. Professor |
| 4. Smt.Sarada Samanta | Stenographer |

VI. Information and Documentation Unit

- | | |
|----------------------------|---------------------------|
| 1. Mr.R.B.Gaddagimath | Librarian, Unit-in-charge |
| 2. Smt.P.V.Nirmala | Cataloguer |
| 3. Smt. K.Swarajya Lakshmi | Jr.Lib.Asstt. |
| 4. Kum.K.Lalitha | Jr.Stenographer |
| 5. Smt.Y.Dhanalakshmi | SSG I |

VII. Centralised Services

(i) Farm Operation Unit

- | | |
|--------------------------|---------------------|
| 1. Mr.V.Murali | Farm Manager |
| 2. Sn.Mohd. Abdul Basith | Lab Technician |
| 3. Sh.K.Rangaswamy | Field Asstt. |
| 4. Sh.B.Veeraiah | Farm Asstt. |
| 5. Sh.P.Neelakantam | Jr.Clerk |
| 6. Sh.K.Thirumalaiah | Pump House Operator |
| 7. Sh.K.Obulapathi | Pump House Operator |

(ii) Production and Publication

- | | |
|---------------------------|---------------|
| 1. Sh.R.V.V.S.Prakasa Rao | Asstt. Editor |
| 2. Sh.A.Murali Mohan Rao | Artist |
| 3. Sh.S.Swamy | SSG III |

(iii) Administration and Accounts

- | | |
|--------------------|---|
| 1. Sh.G.C.Sharma | Sr.Admn.Officer, Unit-in-charge |
| 2. Sh.P.V.R.Murthy | Asstt. Accounts Officer
(retired on 31.8.1986) |

3. Sh.Mohd. Habeebuddin	Superintendent
4. Sh.D.Venkateswarlu	Jr.Hindi Translator
5. Sh.K.Janakiramaiah	Assistant
6. Sh.T.Ashok Kumar	Assistant
7. Sh.Y.Sankar Rao	Sr. Clerk
8. Sh.P.P.Brahmaji	Sr. Clerk
9. Sh.V.Muralikrishna	Jr.Stenographer
10. Sh.M.Venkatesh	Jr.Stenographer
11. Sh.C.Bagaiah	Jr.Clerk
12. Sh.W.Sreenivasa Bhat	Jr.Clerk
13. Sh.M.Narasimha Rao	Jr.Clerk
14. Sh.K.Chandra Mohan	Jr.Clerk (relieved on 14.6.1986)
15. Sh. M.Dinesh	Jr.Clerk
16. Sh.M.J.Pradeep Kumar	Jr.Clerk
17. Sh. Thyag Raju	Jr.Clerk (relieved on 28.11.1986)

Hostel Services

1. Sh. Sham Bahadur	Catering-in-charge
2. Sh.K.Prabhudas	Jr.Clerk
3. Sh.P.G.Kohad	Jr.Clerk (Hindi Typist)
4. Smt.D.Sangamma	Cook (SSG II)
5. Sh.B.Subba Rao	Asstt. Cook (SSG I)
6. Smt.B.Santhamma	SSG I
7. Smt.S.Shakuntala	SSG I (joined on 30.9.1986)

Other Supporting Services

1. Sh.H.S.Iyer	Security Officer-cum-Care-taker (relieved on 23.10.1986)
2. Sh.L.Ramesh	Electrician-cum-plumber
3. Sh.K.Sattaiah	Driver
4. Sh.S.K.Peera	Driver

5. Sh.Mohd.Naseer Khan	Driver
6. Sh.Clyde Pearce	Driver (relieved on 22.11.1986)
7. Smt.G.Mani Bai	SSG II
8. Sh.B.Narayana	SSG II
9. Sh.B.H.Dharma Raj	SSG II
10. Sh.P.Bal Raj	SSG I
11. Sh.Phool Kumar	SSG I
12. Sh. Khalid	SSG I
13. Sh.M.Padmaiah	SSG I
14. Sh.M.Yadaiah	SSG I
15. Sh.G.V.Bikshapathi	SSG I
16. Sh.K.Satyanarayana	SSG I

